Flood Exposure and Asset Repricing: Evidence from FEMA LOMRs and ZIP-Code Home Values

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Abstract

Do coastal housing markets reprice properties when flood-risk changes? I study this question using FEMA Letters of Map Revision (LOMRs), which are localized revisions to effective National Flood Insurance Program (NFIP) flood maps that can change a property's mapped flood-risk designation, floodplain boundary, or related regulatory flood hazard information. In a quarterly ZIP-level event study across U.S. coastal counties from 2009 to 2022, with ZIP and county-by-year fixed effects, I find that home values show near-zero response at impact but decline by approximately 2.8 percent four or more years after a LOMR. The decline is larger in an interaction specification with pre-treatment flood-insurance penetration. This effect is concentrated in ZIP codes located in Democratic-leaning counties, while ZIP codes in Republican-leaning counties show no statistically meaningful response. These findings suggest slow, uneven capitalization of regulatory flood-risk information rather than immediate repricing.

Keywords: Flood risk; property values; FEMA; capitalization; flood insurance; event study

JEL Codes: R31; Q54; G22; H84

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1 Introduction

Flood risk affects housing markets through official designations as well as realized disasters. Federal Emergency Management Agency (FEMA) flood maps assign properties to regulatory flood-risk zones, which can affect insurance requirements, disclosure, underwriting, and buyer beliefs. The most economically salient threshold is often the Special Flood Hazard Area (SFHA), but Letters of Map Revision (LOMRs) can also matter more broadly because they update the official map that private actors use in transactions. I study whether housing markets reprice when those official designations change through FEMA LOMRs.

These stakes are large because the NFIP is both economically important and fiscally strained. As of 2025, the program covers more than 4.7 million policies with roughly $1.3 trillion in insured value, yet it still carries $22.5 billion in debt to the U.S. Treasury after Congress cancelled $16 billion of NFIP debt in 2017 (Congressional Research Service, 2025a,b). Prior work also suggests that flood risk is not fully incorporated into housing prices: Hino and Burke (2021) estimate that floodplain properties are collectively overvalued by about $44 billion and that rezoning into the floodplain is associated with roughly a 2 percent price discount.

These stakes make it economically important to understand how housing markets respond when FEMA revises the official map itself. To study that question, I construct a quarterly ZIP-level panel for coastal U.S. counties from 2009 to 2022 that combines Zillow home values with FEMA LOMRs, National Flood Insurance Program (NFIP) data, and county controls (Federal Emergency Management Agency, 2024a,b; Zillow Research, 2024). A further contribution is the construction of a linked panel that brings together home values, official flood-map revisions, insurance exposure, and local controls in one ZIP-quarter dataset. I run a staggered event study with ZIP and county-by-year fixed effects on first in-window LOMRs. I then interact treatment with pre-treatment NFIP policy penetration

to analyze whether repricing is larger where flood risk is already economically relevant.

Home values do not fall sharply after a LOMR. In the baseline specification, the estimated effect is close to zero at treatment and reaches about 2.8 percent four or more years later. The response is stronger in the continuous insurance-intensity specification, although the quartile pattern is non-monotonic. An exploratory political split also suggests weaker repricing in ZIP codes located in more Republican-leaning counties, which is suggestive of heterogeneity in attention to official flood-risk information. However, the political interaction fails the pre-trend test, so I treat that comparison as descriptive.

I study four related hypotheses. First, LOMRs are associated with lower home values. Second, if housing markets adjust with frictions, that response should emerge gradually after treatment rather than on impact. Third, if flood insurance mainly shields owners from repricing, higher pre-treatment NFIP exposure should weaken the response, while if salience dominates it should strengthen it. Fourth, if political beliefs shape attention to official flood-risk information, repricing should be weaker in more Republican counties. I test the first two with the baseline event study, the third with the intensity and insurance-outcome specifications, and the fourth with a political interaction that I treat as exploratory because its pre-trends are weaker.

Section 2 describes the institutional setting, Section 3 reviews the related literature, and Section 4 and Section 5 present the data and design. Section 6 reports the main results, Section 7 examines insurance-market responses, Section 8 presents exploratory heterogeneity and robustness checks, Section 9 discusses interpretation and limitations, and Section 10 concludes and discusses future research.

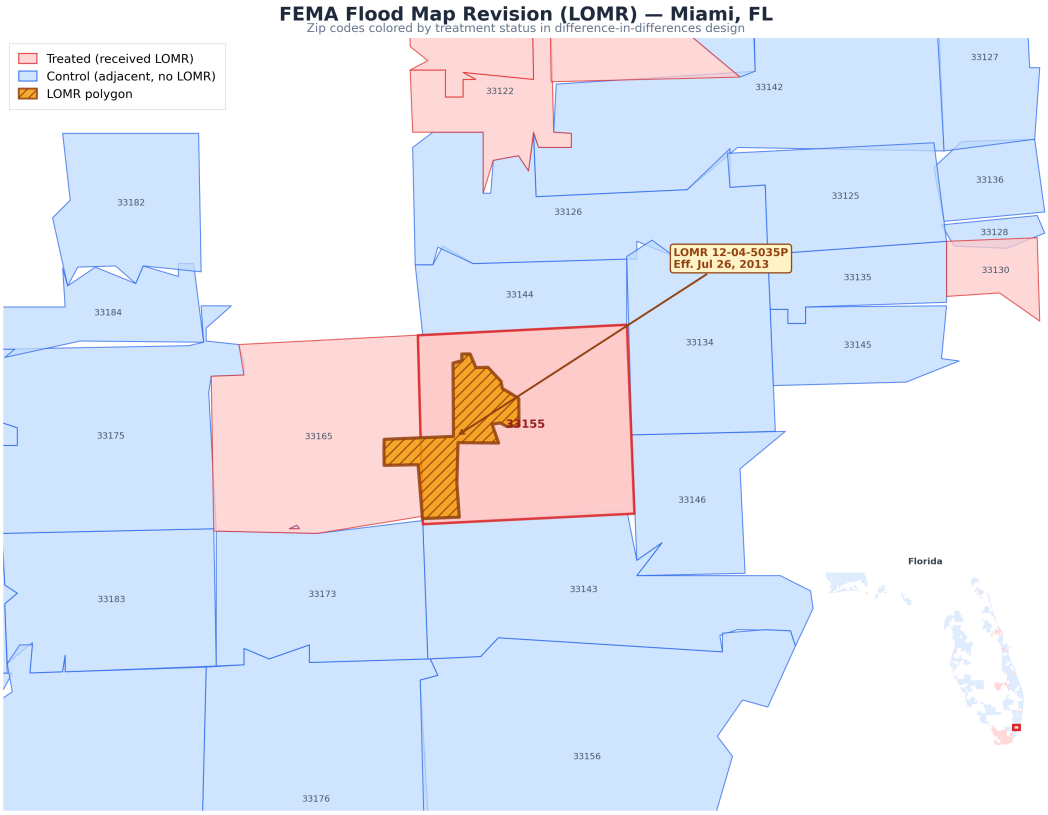
2 Institutional Background

2.1 FEMA Flood Maps and LOMRs

FEMA's Flood Insurance Rate Maps (FIRMs) assign properties to regulatory flood-risk zones and provide related hazard information for lenders, insurers, local governments, and buyers. Letters of Map Revision (LOMRs) update those classifications after new topographic, hydrologic, engineering, or physical information. A LOMR can move land into or out of the Special Flood Hazard Area (SFHA), revise the location of the mapped floodplain boundary, or change other map-based flood hazard information that matters for underwriting, disclosure, and insurance. The SFHA remains the most consequential threshold because federally backed mortgages in participating NFIP communities generally require flood insurance for improved structures inside that zone. However, the broader point for this paper is that LOMRs change the official map that market participants use in transactions, even when the revision does not map cleanly into a simple up-or-down SFHA crossing or follow a recent flood that has already drawn attention to the area.

Figure 1 shows a LOMR in Miami. The revision still makes the treatment structure concrete: the LOMR polygon overlaps two ZIP-codes.

Figure 1: Illustrative LOMR and Treated ZIP Code



Note: Deliberately large illustrative FEMA LOMR in Miami, Florida, chosen for visibility rather than representativeness. The shaded polygon is the revision footprint, the two ZIP-codes (33165 and 33155) are treated in the event-study design, and adjacent ZIP-codes remain untreated in that event window. The inset shows the location within Florida.

2.2 Why LOMRs Should Affect Housing Markets

I expect LOMRs to affect housing prices because they update the official map that market participants use to evaluate expected ownership costs. The most salient case is a revision that changes whether a property is treated as inside the SFHA, but even a boundary revision or other change in mapped flood hazard information can matter if it changes how lenders, insurers, appraisers, or buyers interpret the property. Because those actors use the official map in transactions, a LOMR can have pricing content even when no flood occurs nearby.

For example, suppose a home is under contract when a FEMA LOMR becomes effective. During underwriting, the lender's flood determination may reflect the updated FEMA

map. If the revision changes whether the property is in a Special Flood Hazard Area, the borrower may face new flood-insurance and disclosure requirements before closing. More generally, even when the revision does not change SFHA status, it can still revise the mapped floodplain boundary, base flood elevation, floodway configuration, or other official hazard information that lenders, insurers, appraisers, and buyers use to interpret the property. In that broader sense, a LOMR can change expected ownership costs, documentation, underwriting treatment, or buyer perceptions of risk even without a simple inside-versus-outside SFHA transition. Prices may then adjust gradually as the new information diffuses through transactions, listings, appraisals, and comparable sales rather than through an immediate market-wide reset.

2.3 Why Effects May Be Heterogeneous

Not all revisions should have the same price effect. I expect larger responses where pre-treatment NFIP participation is higher because more households already sit near the margin where insurance costs matter. I also expect larger responses when a revision crosses the SFHA boundary or when disclosure rules make the new status harder to ignore. However, I do not assume that every interaction must point in one direction. Downzoning can reduce required insurance, but it may not remove stigma, and local political environment may matter if households discount official risk information. I treat those extensions as secondary tests rather than main results.

3 Related Literature

3.1 Natural Hazard Risk and Housing Markets

This paper relates first to the literature on natural-hazard risk and housing prices. Early work establishes that flood risk can be capitalized into property values, especially when risk becomes salient. Bin and Polasky (2004) find that homes in the floodplain sell at a discount after Hurricane Floyd. Hallstrom and Smith (2005) demonstrate that even a near-miss hurricane can move housing values by conveying new risk information. Kousky (2010) finds that severe flood experience can change how buyers price floodplain exposure. Related work studies how those discounts evolve over time and across institutional settings. Bin and Landry (2013) document changes in implicit flood-risk premiums, Atreya et al. (2013) study the fading of flood-risk discounts over time, and Beltrán et al. (2018) report a meta-analytic average discount of about 4.6 percent for floodplain properties. More recent work extends this literature from realized disasters to broader climate-risk signals. Hino and Burke (2021) show that regulatory flood-risk information can affect property values even without contemporaneous damage, although aggregate pricing remains incomplete and discounts are larger in more risk-aware markets. Relative to that literature, I focus on FEMA LOMRs, which are localized revisions to the official flood map itself and therefore provide sharper timing variation around changes in regulatory flood-risk information.

3.2 Climate Beliefs, Information, and Disclosure

This paper also relates to work on beliefs, salience, and the transmission of public risk information into asset prices. In the broader climate-risk and real-estate literature, Bernstein et al. (2019) document discounts for sea-level-rise exposure, while Murfin et al. (2020) provide more skeptical evidence on whether those long-run risks are fully priced. Bakkensen and

Barrage (2022), Gibson et al. (2020), and Baldauf et al. (2020) emphasize that capitalization depends in part on beliefs and attention rather than on physical exposure alone. That mechanism is directly relevant here because a LOMR can lower the private cost of learning about mapped flood risk by updating the official information used in transactions.

A complementary strand examines disclosure and information frictions within housing markets. Troy and Romm (2004) and Pope (2008) find larger floodplain discounts under stronger disclosure rules, which is consistent with buyers underreacting when risk information is costly to acquire. Weill (2023) is especially close in spirit because it studies the effects of official FEMA map updates on flood-insurance take-up. My design complements that literature by asking whether localized official map revisions are also capitalized into housing values, and whether that capitalization is stronger where disclosure environments and pre-treatment insurance exposure make the information more salient.

3.3 Difference-in-Differences with Staggered Treatment

Methodologically, the paper uses a staggered difference-in-differences design, so the recent treatment-timing literature matters directly. Goodman-Bacon (2021), Sun and Abraham (2021), and Callaway and Sant'Anna (2021) show why two-way fixed-effects event studies can be misleading when treatment timing and effects vary. Those concerns are relevant here because LOMRs arrive at different dates and the estimated effects appear to build gradually over event time. I therefore emphasize dynamic estimates, interpret pooled post-treatment coefficients cautiously, and report heterogeneity-robust checks as supporting evidence rather than as a separate substantive contribution.

3.4 Positioning

Among prior papers on housing-price effects of flood-zone remapping, Shr and Zipp (2019) is the closest comparison. They study flood-zone remapping and find that prices fall when properties move into the floodplain, while moves out of the floodplain do not generate a symmetric rebound. My paper is also closely related to Weill (2023), which studies official FEMA map updates as information shocks in the flood-insurance market, and to Hino and Burke (2021), which studies how regulatory flood-risk information affects property values at a national scale. I build on that literature by shifting attention to localized FEMA LOMRs, which let me study how markets respond to official map revisions that vary in timing across ZIP codes.

More specifically, the paper contributes to the literature in three ways. First, it studies a different empirical object than most prior flood-risk papers: localized official map revisions rather than realized flood events, broad climate-risk disclosures, or static cross-sectional differences between homes inside and outside the floodplain. Second, the staggered ZIP-level panel allows me to study the timing of capitalization after a revision becomes effective, which helps distinguish gradual repricing from an immediate market-wide reset. Third, by interacting treatment with pre-treatment NFIP penetration, I test whether the same official map change carries more pricing content where flood insurance and regulatory flood designations were already more financially material. I also explore political heterogeneity as a secondary extension, but I interpret those results cautiously because the pre-trend diagnostics are weaker. In that sense, the paper connects the flood-risk literature to a broader literature on how official regulatory reclassification and public information updates enter asset prices.

4 Data

4.1 Core Panel Structure

My main estimation sample is a ZIP-by-quarter panel for U.S. coastal ZIP codes from 2009 to 2022. I begin with a monthly panel, because Zillow home values, NFIP insurance records, and several controls are observed at that frequency. Then, I collapse the data to quarters by averaging the monthly log values within each quarter, because the Zillow Home Value Index (ZHVI) is a three-month smoothed series and monthly observations inherit substantial serial correlation from that construction. After the sample restrictions described below, the quarterly panel contains 4,272 ZIP codes.

Those 4,272 ZIP codes include 398 single-LOMR treated ZIP codes within the estimation window, 3,711 never-treated ZIP codes, and 163 additional ZIP codes whose first LOMR occurs after the 2009–2022 sample window. I retain those 163 ZIP codes because they are not yet treated during the sample period and can still contribute control observations. These counts differ from the earlier geographic sample in Table 1, which reports the wider coastal universe before the event-study restrictions.

4.2 Data Sources

The primary outcome is Zillow's ZIP-level Home Value Index (ZHVI), using the all-homes (single-family residences plus condo/co-op), middle-tier, smoothed, seasonally adjusted series, which I deflate to December 2022 dollars using the Consumer Price Index for All Urban Consumers (CPI-U) (Federal Reserve Bank of St. Louis, 2024) and transform into ln(real ZHVI) for the main specifications. Zillow's coverage determines the first major sample screen because a nontrivial share of coastal ZIP codes do not have usable home-value data over the full period.

Treatment comes from FEMA's S_LOMR polygon layer in the National Flood Hazard Layer (Federal Emergency Management Agency, 2024a). I spatially overlay those polygons onto Census ZCTA boundaries (U.S. Census Bureau, 2024) to identify whether a ZIP code is ever exposed to a LOMR and whether the first effective LOMR falls inside the 2009–2022 analysis window. The overlay supports the geographic-intensity extension directly. For the zoning-direction analysis, I use post-LOMR changes in SFHA share, defined as the fraction of a ZIP code's NFIP policies whose rated flood zone falls in the SFHA (A or V zones) in a given month, as a proxy for whether local flood exposure shifts upward or downward.

I merge those treatment data with OpenFEMA NFIP policy and claims records (Federal Emergency Management Agency, 2024b), which I aggregate to the ZIP-quarter level. The insurance data provide policy counts, average premiums, the share of policies rated in SFHA zones, and claims activity. Those variables serve two purposes. They measure pre-treatment exposure to flood insurance, and they also provide outcomes for the mechanism tests. From the policy counts I also construct *policy intensity*, defined as the pre-LOMR average NFIP policy count divided by ZIP population, which serves as the main treatment-interaction variable in the intensity specification.

I add county-level unemployment rates from BLS Local Area Unemployment Statistics (LAUS) (U.S. Bureau of Labor Statistics, 2024) as a local macro control, and I merge county presidential vote returns from the MIT Election Data and Science Lab (MIT Election Data and Science Lab, 2017) to build the political heterogeneity measure. Finally, I hand-code state flood-disclosure regimes into strict and broad indicators. That combination gives me a panel that links local home values, official flood-map revisions, insurance exposure, and institutional context in a common ZIP-quarter frame.

4.3 Sample Construction

Table 1 summarizes the sample construction. I start from 33,782 U.S. ZIP codes, restrict to 5,450 ZIP codes in NOAA coastal counties, and then merge Zillow coverage, which leaves 4,836 coastal ZIP codes with usable home-value data. The LOMR–ZCTA overlay identifies 1,125 ever-treated ZIP codes in that sample, including 855 whose first observed LOMR falls inside the 2009–2022 window.

Then, I impose the event-study restrictions. First, I drop ZIP codes that were already treated before 2009 because they do not have a clean pre-period. Second, I drop ZIP codes with multiple LOMRs because overlapping revisions make event time difficult to interpret. I also winsorize ln(real ZHVI) at the 1st and 99th percentiles to limit the influence of extreme observations. After these restrictions, the final regression panel contains 4,272 ZIP codes. I impose these restrictions because clean treatment timing has more value here than the additional observations they would preserve.

4.4 Summary Statistics

Table 2 reports the main descriptive moments for the estimation sample, organized by variable role. Mean real home value is $486,215, while the median is $374,153, which indicates the expected right tail in coastal housing markets. Among the controls, the average ZIP-quarter has 65.21 NFIP policies and an average premium of $710.36. Claims are much more infrequent: the median ZIP-quarter records zero claims, even though the mean is 1.13. That pattern matters because it reinforces the distinction between map revisions and realized damage events. The average SFHA zone share across all coastal ZIP codes is 0.35, which means that about one-third of NFIP policies in the typical ZIP are rated in a SFHA.

The treatment variables are also highly skewed. The mean policy-intensity measure is 5.93 policies per 1,000 residents, while the median is only 0.68, and treatment intensity has

Table 1: Sample Construction: From US Zip Codes to Estimation Panel

Step	Description	Zip Codes
Panel A: Geographic sample		
1	All US zip codes	33,782
2	Restrict to NOAA coastal counties (ocean-bordering)	5,450
3	Merge with Zillow ZHVI (drop zips without coverage)	4,836
Panel B: Treatment classification (within 4,836)		
	LOMR–ZCTA spatial overlay pairs	2,936
	Zip codes with $\geq$1 effective LOMR	1,125
	First LOMR in window (all treated zips)	855
	Single LOMR, first in window (estimation treated group)	398
	Multi-LOMR zips	550
	Already treated before 2009	54
	First LOMR after sample window	216
	Never-treated zip codes	3,711
Panel C: Event-study restrictions		
4	Drop already-treated (no clean pre-period)	−54
	Drop multi-LOMR (overlapping treatment timing)	−510
	Final estimation sample	**4,272**
	Single-LOMR treated (in window)	398
	Not-yet-treated (first LOMR after window)	163
	Never-treated controls	3,711
5	Quarterly panel (after collapse)	228,005 obs

Notes: Sample begins with all US zip codes from the SimpleMaps database. Step 2 restricts to zip codes within NOAA-designated coastal counties that border the ocean (excluding Alaska, Hawaii, and US territories). Step 3 drops zips lacking Zillow Home Value Index coverage. Treatment is defined by spatial overlay of FEMA LOMR polygons onto Census ZCTA boundaries. Categories in Panel B are not mutually exclusive: a zip can be both multi-LOMR and already-treated. Panel C drops are applied sequentially; the multi-LOMR drop count (510) is net of the 54 already-treated zips removed in the prior step. Not-yet-treated zips (163) retain single LOMRs arriving after 2022 and contribute control observations during the sample period. The quarterly panel collapses monthly observations by averaging log values within each quarter.

a median of zero. Therefore, most coastal ZIP codes have limited flood-insurance exposure, while a smaller subset is much more exposed. Those patterns still shape the empirical strategy, because they suggest that average treatment effects may mask larger responses in places where flood insurance is already economically meaningful.

Table 2: Summary Statistics: Estimation Sample (2009–2022)

	count	mean	sd	min	p25	p50	p75	max
Panel A: Outcomes								
Home Value Index (Dec 2022 $)	228,005	486,214.62	416,676.37	32,424.71	234,258.39	374,152.97	594,862.94	8,125,586.00
ln(Real ZHVI)	228,005	12.85	0.67	11.36	12.36	12.83	13.30	14.56
ln(NFIP Policies + 1)	228,005	2.46	1.80	0.00	0.98	2.08	3.71	7.87
ln(NFIP Claims + 1)	228,005	0.11	0.46	0.00	0.00	0.00	0.00	8.16
Panel B: Treatment Variables								
Post-LOMR	228,005	0.04	0.19	0.00	0.00	0.00	0.00	1.00
Ever Treated	228,005	0.13	0.34	0.00	0.00	0.00	0.00	1.00
Policy Intensity (per 1,000)	228,005	5.93	21.07	0.00	0.18	0.68	4.13	716.99
Treatment Intensity (per 1,000)	228,005	3.60	32.54	0.00	0.00	0.00	0.00	944.01
Upzoned (into SFHA)	228,005	0.05	0.21	0.00	0.00	0.00	0.00	1.00
Downzoned (out of SFHA)	228,005	0.03	0.18	0.00	0.00	0.00	0.00	1.00
SFHA-Crossing LOMR	228,005	0.08	0.27	0.00	0.00	0.00	0.00	1.00
Panel C: Controls								
County Unemp. Rate (NFIP Policies (qtr avg)	228,005	65.21	164.11	0.00	1.67	7.00	39.67	2,624.67
NFIP Avg Premium ($)	228,005	710.36	631.82	0.00	357.67	565.67	914.10	19,570.33
NFIP Claims (qtr avg)	228,005	1.13	25.22	0.00	0.00	0.00	0.00	3,498.00
Panel D: Heterogeneity Measures								
Republican County	222,846	0.50	0.50	0.00	0.00	1.00	1.00	1.00
R Two-Party Vote Share	222,846	0.42	0.15	0.06	0.32	0.43	0.52	0.89
Disclosure: Strict (9 states)	228,005	0.29	0.45	0.00	0.00	0.00	1.00	1.00
Disclosure: Broad (13 states)	228,005	0.61	0.49	0.00	0.00	1.00	1.00	1.00
Panel E: Sample Characteristics								
Zip Population	228,005	19,717.91	17,846.46	25.00	5,236.00	15,229.00	29,309.00	109,931.00
Zip Pop. Density	228,005	4,960.31	11,251.76	0.90	225.60	1,571.80	4,837.00	143,070.70
SFHA Zone Share	228,005	0.35	0.31	0.00	0.00	0.33	0.60	1.00

4.5 Balance Table

Table 3 compares pre-treatment means for the main outcome, the insurance variables, and the local controls across treated ZIP codes and never-treated controls. Pre-treatment home values are very similar across the two groups, while treated ZIP codes show higher NFIP exposure and somewhat different local characteristics. That pattern reinforces the case for relying on within-ZIP timing variation with ZIP fixed effects rather than a cross-sectional comparison.

Table 3: Balance Table: Pre-Treatment Characteristics

	Control	Treated	Difference
Home Value (Dec 2022 $)	478,848.23	474,199.50	4,648.72
Population	18,542.78	21,918.10	-3,375.32***
Pop. Density (per sq mi)	5,027.05	3,123.25	1,903.80***
County Unemp. Rate (NFIP Policies (qtr avg)	59.98	84.55	-24.57***
NFIP Avg Premium ($)	690.42	781.76	-91.35***
SFHA Zone Share	0.34	0.40	-0.06***
NFIP Claims (qtr avg)	1.07	1.35	-0.28

Treated: zips with single LOMR during 2009-2022.
Control: zips with no LOMR.
Pre-treatment means reported for treated zips.
Difference = Treated - Control. Welch t-test.

5 Empirical Strategy

5.1 Baseline Event-Study Specification

I estimate dynamic effects with a staggered event-study difference-in-differences design because the theory predicts gradual capitalization after a LOMR rather than an immediate one-quarter jump. The unit of observation is the ZIP-quarter. I assign treatment timing with the first effective LOMR in the 2009–2022 window, and I exclude ZIP codes with earlier treatment or multiple LOMRs because event time is otherwise difficult to interpret.

$$\ln(\text{Real ZHVI})_{z,t} = \alpha_z + \delta_{c(z),y(t)} + \sum_{\tau \neq -1} \beta_\tau \cdot \mathbf{1}[t - E_z = \tau] + \gamma' \mathbf{X}_{z,t} + \varepsilon_{z,t} \tag{1}$$

In Equation 1, α_z denotes ZIP fixed effects, $\delta_{c(z),y(t)}$ denotes county-by-year fixed effects, and E_z is the effective date of the first LOMR for ZIP z. The coefficients of interest, β_τ, compare treated ZIP codes at each event-time bin to the omitted year immediately before treatment, $\tau = -1$. I group event time into nine annual bins from $\tau = -4$ through $\tau = +4$: an endpoint bin for three or more years before treatment, two interior pre-treatment bins, the omitted reference year immediately before treatment, four post-treatment year bins, and an endpoint bin for four or more years after treatment. The main control vector $\mathbf{X}_{z,t}$ includes

county unemployment and contemporaneous NFIP policy counts. I winsorize ln(real ZHVI) at the 1st and 99th percentiles to limit the influence of outliers. I weight observations by ZIP population and cluster standard errors at the county level because treatment timing and housing-market shocks are unlikely to be independent within counties.

I prefer this specification to a single post-treatment indicator because a pooled TWFE coefficient would average together near-zero early effects and more negative late effects. That compression has a real cost. It hides the timing pattern that matters for the capitalization question, and it makes the main estimate harder to interpret economically.

5.2 Intensity Specification

I next ask whether the same map revision has a larger price effect where flood-insurance exposure was already more salient before treatment. The relevant margin here is not the size of the LOMR polygon itself. It is the extent to which households in the ZIP already sat close to the insurance and disclosure margin before the map change arrived.

$$\ln(\text{Real ZHVI})_{z,t} = \alpha_z + \delta_{c(z),y(t)} + \sum_{\tau \neq -1} \beta_\tau \cdot \mathbf{1}[t - E_z = \tau] \times \underbrace{\frac{\text{Policies}_z^{\text{pre}}}{\text{Pop}_z}}_{\text{Intensity}_z} + \gamma' \mathbf{X}_{z,t} + \varepsilon_{z,t} \quad (2)$$

Here Intensity_z is the ZIP's pre-LOMR NFIP policy count per resident. I construct it from pre-treatment policy levels so the interaction captures pre-existing exposure rather than post-treatment behavioral adjustment. That distinction matters because I want the interaction to proxy for salience, not to fold an outcome of the treatment back into the treatment itself. The coefficients in Equation 2 therefore show how the dynamic effect varies with baseline insurance penetration. I interpret those coefficients at realistic values of intensity, such as the sample mean or the quartiles, because an intensity value of one lies far from the center

of the data.

5.3 Identification

Identification comes from comparing within-ZIP changes in home values around the first LOMR to changes in never-treated and not-yet-treated ZIP codes observed in the same county-year. ZIP fixed effects absorb time-invariant differences across local markets, such as baseline housing quality, persistent amenity value, and long-run flood exposure. County-by-year fixed effects absorb annual shocks that operate at the county level, including broad labor-market conditions and county housing cycles.

The key assumption is that, absent the LOMR, treated ZIP codes would have followed the same trend as comparison ZIP codes within the same county-year cell. I cannot test that assumption directly, and I do not want to state it too strongly. A LOMR can follow local development, drainage work, or engineering review, and those processes may coincide with housing-market changes for reasons other than flood-risk information. That concern does not invalidate the design, but it remains the main threat to identification. I address it with the sample restrictions above, rich fixed effects, and explicit pre-trend tests. I also report a Callaway and Sant'Anna check later because staggered timing can distort TWFE event studies when treatment effects vary across cohorts or over event time.

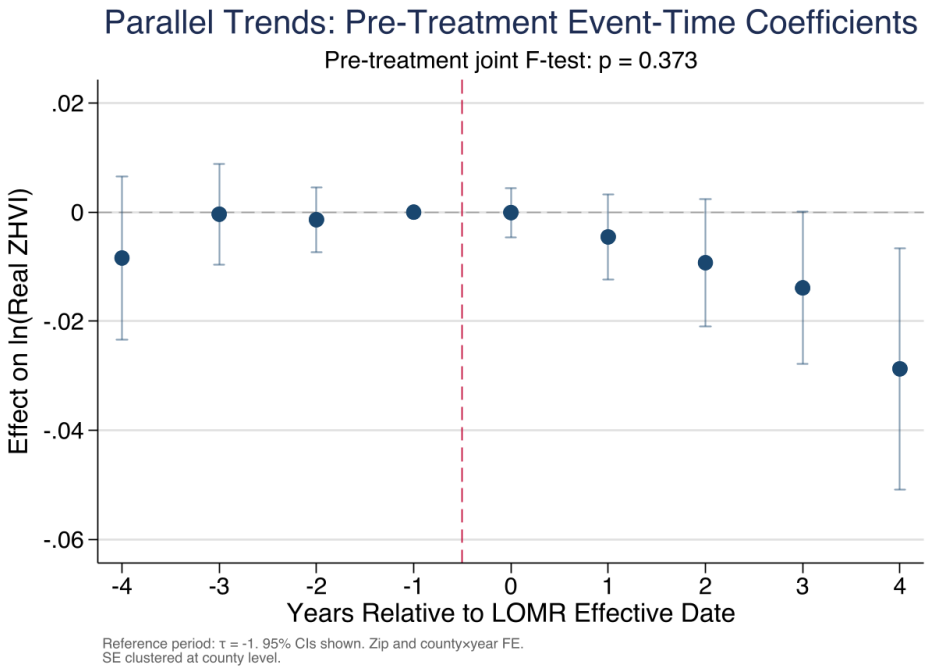
5.4 Parallel Trends

The most direct diagnostic is the behavior of the lead coefficients (Figure 2). If treated ZIP codes were already on a different trend before the LOMR, the post-treatment profile would be harder to interpret as capitalization around the revision itself. In the baseline specification, the three pre-treatment coefficients are jointly insignificant, with $F(3, 338) = 1.04$ and $p = 0.373$. Therefore, the pre-treatment path is consistent with parallel trends at

the annual-bin level.

However, I treat that result as necessary evidence, not a guarantee. The test has limited power, and it cannot rule out slow-moving local shocks that line up with the engineering or administrative process behind a map revision. I read the pre-trend evidence as supportive because it reduces one obvious concern, but it does not eliminate the broader identification problem on its own.

Figure 2: Parallel Trends: Pre-Treatment Event-Time Coefficients



Note: Event-time coefficients from the main specification (Equation 1) with 95% confidence intervals. Reference period: $\tau = -1$ (12–0 months before LOMR). ZIP and county-by-year fixed effects. Standard errors clustered at county level (339 clusters).

6 Main Results

6.1 Baseline Effects of LOMRs

Table 4 reports the main test of H1 and H2 through baseline event-study regressions for ln(real ZHVI). The dependent variable is log real ZIP-level home value, and the key regressors are event-time indicators for years relative to the first in-window LOMR. Across columns, I move from no controls to the main control set and then to the fuller insurance-control set, so the table shows whether the treatment path is sensitive to those additions. Figure 3 plots the same coefficients as a dynamic path. The three lead coefficients are small, and the joint pre-trend test from Sub-section 5.4 does not reject. The first two post-treatment bins are also close to zero and statistically imprecise. However, the path becomes more negative over time. In the main specification, the coefficient at three to four years after treatment is -0.0139 with a standard error of 0.0071 and is marginal at the 10 percent level. The coefficient at four or more years after treatment is -0.0287 with a standard error of 0.0113, a 95 percent confidence interval of $[-0.0508, -0.0066]$, and a p-value of 0.011.

Economically, the late-post-treatment estimate is associated with about a 2.8 percent decline in real home values relative to the year before treatment. At the sample mean real home value in the estimation sample, that magnitude corresponds to roughly $14,000. I do not read that as a large immediate shock. I read it as a gradual repricing that accumulates after the map revision enters transactions, underwriting, and buyer expectations.

The estimates also move very little across columns. The $\tau+4$ coefficient ranges from -0.0285 without controls to -0.0287 with the main controls and -0.0287 again with the fuller insurance-control set. That stability matters because it suggests the late negative effect is not especially sensitive to whether I control only for unemployment and NFIP policies or also add premiums and claims.

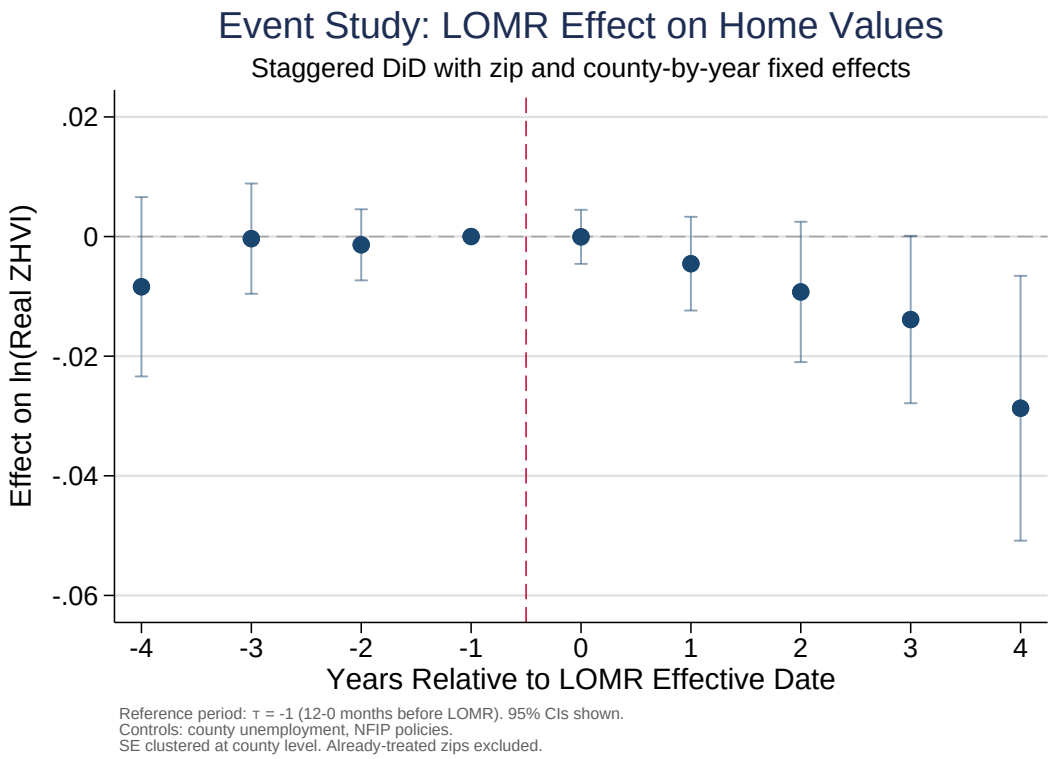
Table 4: ln(Real ZHVI) on LOMR Treatment

	(1) No Controls	(2) Main Controls	(3) Full Controls
τ = -4 (3+ yrs pre)	-0.00840	-0.00839	-0.00836
	(0.00765)	(0.00765)	(0.00765)
τ = -3 (2-3 yrs pre)	-0.000235	-0.000353	-0.000299
	(0.00472)	(0.00470)	(0.00470)
τ = -2 (1-2 yrs pre)	-0.00131	-0.00138	-0.00134
	(0.00304)	(0.00303)	(0.00303)
τ = 0 (0-1 yr post)	0.0000564	-0.0000485	-0.0000269
	(0.00230)	(0.00231)	(0.00231)
τ = +1 (1-2 yrs post)	-0.00435	-0.00454	-0.00457
	(0.00397)	(0.00400)	(0.00399)
τ = +2 (2-3 yrs post)	-0.00908	-0.00926	-0.00923
	(0.00595)	(0.00599)	(0.00599)
τ = +3 (3-4 yrs post)	-0.0139*	-0.0139*	-0.0139*
	(0.00717)	(0.00714)	(0.00714)
τ = +4 (4+ yrs post)	-0.0285**	-0.0287**	-0.0287**
	(0.0113)	(0.0113)	(0.0113)
Observations	228,002	228,002	228,002
Within R^2	0.0015	0.0029	0.0030

Standard errors in parentheses
Zip and county×year fixed effects.
Standard errors clustered at county level.
Reference period: 12-0 months before LOMR (τ = -1).
Already-treated zips (LOMR before 2009) excluded.
(1) No controls. (2) Unemployment rate, NFIP policies. (3) Adds avg premium, claims.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

I interpret this pattern as gradual capitalization rather than sharp repricing. That institutional setting still shapes the timing here. For example, a LOMR may first matter for a small set of pending sales. A lender orders a flood determination, the file reflects the updated FEMA map, and the buyer learns that the property's mapped flood status or nearby official hazard information has changed. That update may alter expected insurance costs, documentation requirements, underwriting treatment, or the buyer's perception of risk, and the seller may adjust after that transaction or after losing that buyer. Later transactions then carry the information further through listings, appraisals, and comparable sales. Because

Figure 3: Event Study: LOMR Effect on Home Values



Note: Coefficient estimates from the main specification (Equation 1) with 95% confidence intervals. Reference period: $\tau = -1$. Controls: county unemployment, NFIP policies. ZIP and county-by-year fixed effects. Standard errors clustered at county level. Population-weighted.

ZIP-level home values reflect a sequence of transactions rather than a single market-clearing price, I would expect a slower adjustment path.

However, I do not want to overstate the claim. The event-study profile is consistent with gradual repricing after official flood-risk revisions, but I cannot rule out coincident local changes tied to development, drainage work, or engineering review. Still, the evidence points in one direction: there is little sign of differential decline before treatment, and the negative effects arrive later rather than earlier. Therefore, the baseline results fit a story in which FEMA map revisions are incorporated into housing prices over time instead of on impact.

6.2 Heterogeneity by Pre-Treatment Insurance Exposure

Table 5 turns to H3 and keeps ln(real ZHVI) as the dependent variable while comparing two sets of regressors. Column 1 repeats the binary event-time indicators from the baseline design, and Column 2 replaces them with event-time indicators interacted with pre-treatment NFIP policy intensity. Figure 4 plots the interaction coefficients as a dynamic path. Together they test whether the average treatment effect masks larger declines in places where flood insurance was already more prominent before treatment. In the continuous interaction specification, the later post-treatment coefficients are more negative than in the baseline binary model. The estimate at four or more years after treatment is $\hat{\beta}_{\tau+4} = -7.15^{***}$ with a standard error of 2.54. Evaluated at the sample mean intensity of 0.0059, that coefficient implies about a 4.2 percent decline in real home values, roughly 1.5 times the baseline binary estimate. The joint pre-trend test yields $F(3, 338) = 1.47$ with $p = 0.223$, which is broadly supportive, although I treat the interaction magnitudes with more caution than the baseline event study because the heterogeneous-effects design is less stable overall.

Table 6 uses the same dependent variable, ln(real ZHVI), but now the key regressors are quartile-specific event-time interactions rather than one continuous intensity interaction. Each column corresponds to a different quartile of pre-treatment NFIP policy intensity among treated ZIP codes, so the table shows how the dynamic treatment path differs across low- and high-exposure places. Figure 5 plots those paths directly. This design shows the same idea in quartiles rather than as a continuous interaction. The quartile specification does not show a smooth monotonic gradient. The lowest-exposure quartile is close to zero throughout. The second quartile shows the clearest negative response, with statistically significant coefficients at τ+2, τ+3, and τ+4. The upper quartiles are generally negative, but the estimates are imprecise. I treat that pattern as informative because it suggests that

Table 5: ln(Real ZHVI) on LOMR × Policy Intensity

	(1) Binary Treatment	(2) Intensity-Weighted
τ = -4 (3+ yrs pre)	-0.00839	
	(0.00765)	
τ = -3 (2-3 yrs pre)	-0.000353	
	(0.00470)	
τ = -2 (1-2 yrs pre)	-0.00138	
	(0.00303)	
τ = 0 (0-1 yr post)	-0.0000485	
	(0.00231)	
τ = +1 (1-2 yrs post)	-0.00454	
	(0.00400)	
τ = +2 (2-3 yrs post)	-0.00926	
	(0.00599)	
τ = +3 (3-4 yrs post)	-0.0139*	
	(0.00714)	
τ = +4 (4+ yrs post)	-0.0287**	
	(0.0113)	
τ = -4 × intensity		0.411
		(0.574)
τ = -3 × intensity		-0.130
		(0.279)
τ = -2 × intensity		-0.302
		(0.194)
τ = 0 × intensity		-0.00919
		(0.287)
τ = +1 × intensity		-0.811**
		(0.398)
τ = +2 × intensity		-1.651*
		(0.881)
τ = +3 × intensity		-3.288***
		(1.030)
τ = +4 × intensity		-7.152***
		(2.541)
Observations	228,002	228,002
Within R^2	0.0029	0.0039

Standard errors in parentheses

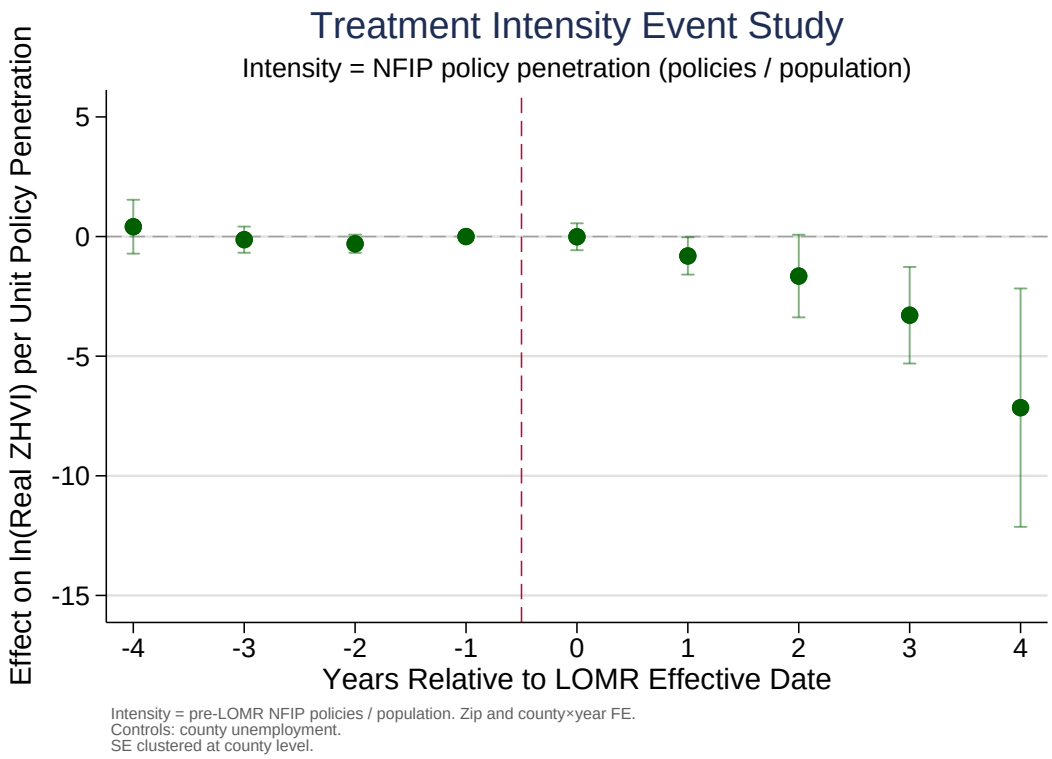
Zip and county×year FE. SE clustered at county level.

Intensity = pre-LOMR NFIP policies / population.

Proxy for share of housing stock in flood zone.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 4: Intensity-Weighted Event Study



Note: Coefficient estimates from the intensity specification (Equation 2) with 95% confidence intervals. Treatment intensity = pre-LOMR NFIP policies per capita.

repricing appears once a ZIP reaches a moderate level of flood-insurance exposure. However, it does not support the stronger claim that each additional increase in exposure produces a larger decline.

I interpret the intensity results as evidence that pre-treatment insurance penetration amplifies the price effect of a LOMR because it proxies for flood-risk salience rather than protection from repricing. If flood insurance mainly insulated owners from revised risk, higher NFIP penetration should weaken the housing-price response. I find the opposite pattern in the continuous interaction specification. That result is consistent with Wagner (2022), who shows that flood-insurance participation reflects pre-existing observable differences related to flood-risk exposure and adaptation rather than simply muting price responses. In my

Table 6: ln(Real ZHVI) by Policy Intensity Quartile

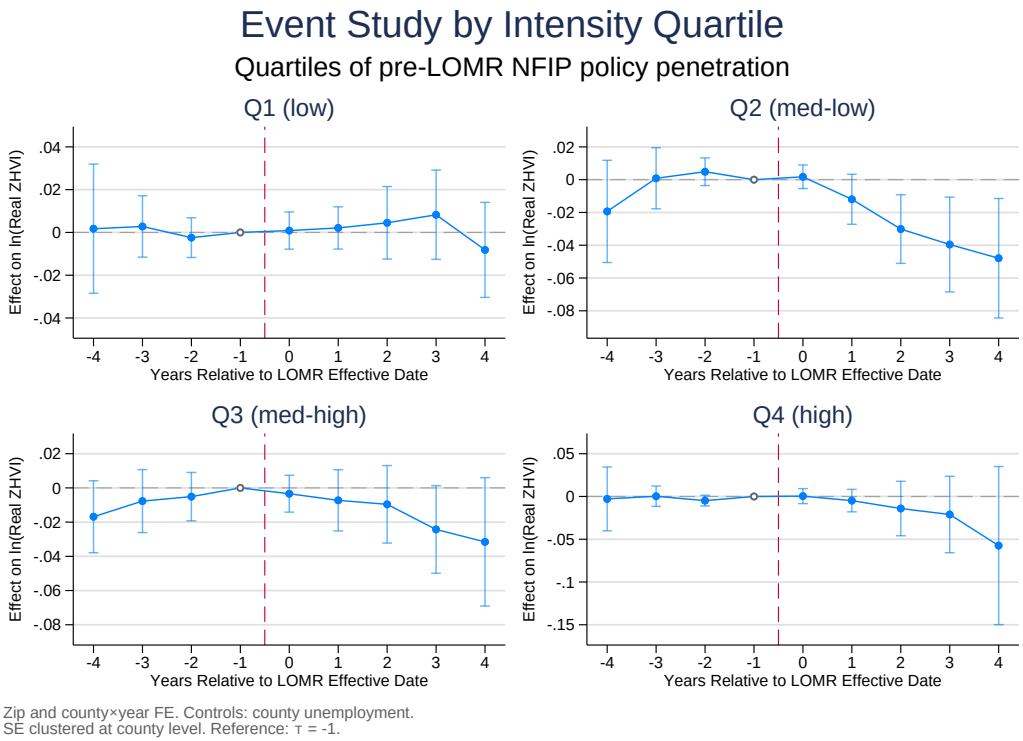
	(1) Q1 (Low)	(2) Q2	(3) Q3	(4) Q4 (High)
τ = -4 (3+ yrs pre)	0.00174	-0.0194	-0.0169	-0.00284
	(0.0154)	(0.0159)	(0.0107)	(0.0190)
τ = -3 (2-3 yrs pre)	0.00279	0.000847	-0.00771	0.000271
	(0.00732)	(0.00950)	(0.00939)	(0.00605)
τ = -2 (1-2 yrs pre)	-0.00243	0.00484	-0.00509	-0.00487
	(0.00474)	(0.00431)	(0.00721)	(0.00318)
τ = 0 (0-1 yr post)	0.000878	0.00176	-0.00339	0.000415
	(0.00444)	(0.00368)	(0.00550)	(0.00448)
τ = +1 (1-2 yrs post)	0.00210	-0.0120	-0.00726	-0.00484
	(0.00505)	(0.00779)	(0.00912)	(0.00671)
τ = +2 (2-3 yrs post)	0.00451	-0.0301***	-0.00960	-0.0141
	(0.00866)	(0.0107)	(0.0116)	(0.0163)
τ = +3 (3-4 yrs post)	0.00826	-0.0396***	-0.0243*	-0.0211
	(0.0107)	(0.0147)	(0.0131)	(0.0228)
τ = +4 (4+ yrs post)	-0.00817	-0.0479**	-0.0315	-0.0574
	(0.0113)	(0.0186)	(0.0191)	(0.0472)
Observations		228,002		
Within R^2		0.0039		

Standard errors in parentheses
Zip and county×year FE. SE clustered at county level.
Quartiles of pre-LOMR NFIP policy penetration among treated zips.
Q1 = lowest exposure, Q4 = highest exposure.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

setting, places with higher pre-treatment insurance penetration may also be places where official flood-risk updates are more salient, so a map revision carries more marginal pricing content. It is less consistent with a pure moral hazard interpretation, where insurance coverage would dampen repricing.

I do not interpret the quartile specification as a clean dose response. The null result in Q1 is informative because it suggests that LOMRs carry little pricing content where flood-insurance exposure is negligible. The fact that only Q2 is statistically significant points instead to a threshold or salience effect. Once exposure reaches a moderate level, repricing becomes detectable. However, the absence of statistical significance in Q3 and Q4 means I

Figure 5: Event Study by Insurance Penetration Quartile



Note: Separate event-study coefficients by pre-LOMR policy intensity quartile. Q1 = lowest penetration, Q4 = highest. 95% confidence intervals shown.

cannot claim that the response increases smoothly with insurance penetration. That pattern may reflect limited precision after splitting the sample, coarse quartile cutoffs, or real nonlinearity in how households process official flood-risk revisions. Therefore, I treat the continuous intensity specification as the main evidence and the quartile design as a descriptive check.

7 Insurance Market Responses

Table 7 provides supporting evidence for H3 rather than a separate headline test. The dependent variable shifts away from home values and onto insurance outcomes. Column 1 uses ln(NFIP policies + 1) as the dependent variable, Column 2 uses ln(NFIP claims + 1), and the key regressors remain the event-time indicators for years relative to the first

LOMR. Figure 6 plots the policy coefficients from Column 1. These results help distinguish salience from realized-damage or immediate insurance-adjustment stories. If LOMRs reduce housing values through immediate insurance adjustment, NFIP policy counts should move around the revision itself. In the aggregate policy event study, I see limited movement in ln(NFIP policies + 1), with at most a brief decline in the early post-treatment window. Claims provide an even cleaner falsification outcome. A LOMR does not generate flooding on its own, and I find no significant post-treatment effects on ln(NFIP claims + 1). That null matters because it is consistent with an informational or regulatory shock rather than a realized disaster shock.

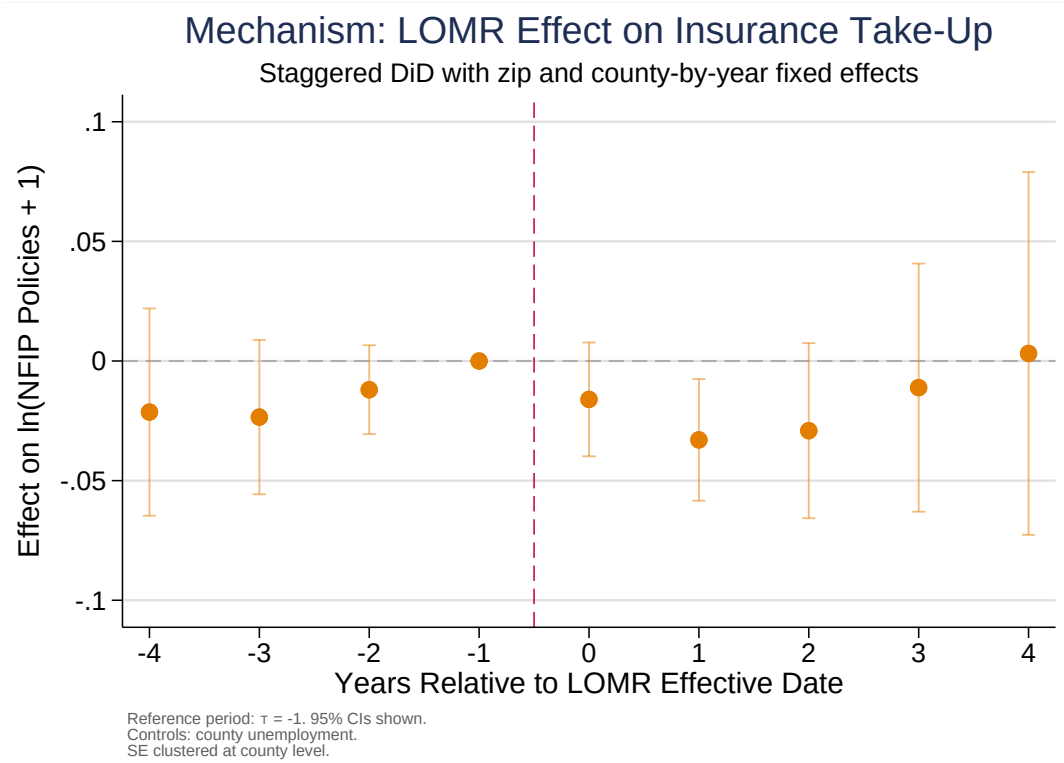
Table 7: ln(NFIP Policies) on LOMR Treatment

	(1) ln(Policies)	(2) ln(Claims)
τ = -4 (3+ yrs pre)	-0.0213	-0.0262*
	(0.0221)	(0.0152)
τ = -3 (2-3 yrs pre)	-0.0234	-0.0286
	(0.0165)	(0.0254)
τ = -2 (1-2 yrs pre)	-0.0120	-0.0177
	(0.00947)	(0.0181)
τ = 0 (0-1 yr post)	-0.0160	0.00154
	(0.0121)	(0.0195)
τ = +1 (1-2 yrs post)	-0.0330**	0.00506
	(0.0130)	(0.0235)
τ = +2 (2-3 yrs post)	-0.0291	-0.0348*
	(0.0187)	(0.0191)
τ = +3 (3-4 yrs post)	-0.0111	-0.0296
	(0.0264)	(0.0232)
τ = +4 (4+ yrs post)	0.00315	-0.0275
	(0.0387)	(0.0226)
Observations	228,002	228,002
Within R^2	0.0021	0.0002

Standard errors in parentheses
Zip and county×year FE. SE clustered at county level.
Policies: NFIP policy count (mechanism). Claims: falsification.
Reference period: τ = -1 (12-0 months before LOMR).
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Splitting the policy response by upzoning and downzoning suggests some decline in downzoned areas, while upzoned areas do not show a comparably strong increase. I interpret that asymmetry as limited mechanism evidence rather than a direct explanation of the home-value estimates, because compliance can be slow and some buyers may respond to new map status before they actually purchase a policy.

Figure 6: Insurance Policies Event Study



Note: Event-study coefficients for ln(NFIP policies + 1) as the dependent variable. Same specification as Equation 1.

8 Exploratory Heterogeneity and Robustness

8.1 Exploratory Heterogeneity

Section 8 turns to H4 and to supporting heterogeneity checks that I treat as exploratory. The overall panel remains large, but several splits leave only about 100 to 225 treated ZIP codes, which reduces precision, and some diagnostics weaken. Table A1 reports the upzoning and downzoning split, and Figure A2 plots the corresponding event-study paths. These results do not produce individually precise home-value effects. The estimate at $\tau+4$ is -0.014 for upzoned ZIP codes with a standard error of 0.017, and -0.024 for downzoned ZIP codes with a standard error of 0.019. Table A2 reports the disclosure-law interactions, and Figure A3 plots the same comparison visually. Those interaction terms are also insignificant. That null is consistent with the LOMR itself being the relevant information event in many transactions, although I cannot rule out low power because the strict-disclosure sample is limited.

Table 8 returns to ln(real ZHVI) as the dependent variable, but the key regressors now split the intensity-weighted event-study coefficients into a baseline path for Democratic-leaning counties and a differential interaction for Republican-leaning counties. Figure 7 plots the implied event-study paths.

These results present a sharper initial contrast on political heterogeneity. Democratic-leaning counties show a large negative baseline intensity effect, while the Republican interaction term is positive and similar in magnitude, which implies a near-zero net effect in more Republican counties. That pattern is consistent with belief heterogeneity dampening the capitalization of official flood-risk information. However, the pre-treatment interaction test rejects parallel trends, with $F(3, 331) = 12.05$ and $p < 0.001$, so I cannot rule out that the partisan gap reflects pre-existing differential trends rather than a differential response to the

Table 8: ln(Real ZHVI) on Policy Intensity × Political Lean

	(1) Intensity × Republican Interaction
τ = -4 × intensity	-3.080*
	(1.760)
τ = -3 × intensity	1.885***
	(0.447)
τ = -2 × intensity	0.877**
	(0.368)
τ = 0 × intensity	-0.979***
	(0.240)
τ = +1 × intensity	-2.937***
	(0.687)
τ = +2 × intensity	-5.640***
	(1.681)
τ = +3 × intensity	-7.046***
	(1.862)
τ = +4 × intensity	-12.72***
	(1.152)
τ = -4 × intensity × Republican	3.942**
	(1.826)
τ = -3 × intensity × Republican	-2.181***
	(0.531)
τ = -2 × intensity × Republican	-1.308***
	(0.403)
τ = 0 × intensity × Republican	1.092***
	(0.389)
τ = +1 × intensity × Republican	2.672***
	(0.811)
τ = +2 × intensity × Republican	5.981***
	(1.789)
τ = +3 × intensity × Republican	6.610***
	(2.201)
τ = +4 × intensity × Republican	11.33***
	(1.997)
Observations	222,843
Within R^2	0.0058

Standard errors in parentheses

Sample: all treated zips + never-treated controls (full event study sample).

ibin = intensity LOMR effect (Dem-leaning counties). rep = differential Republican effect.

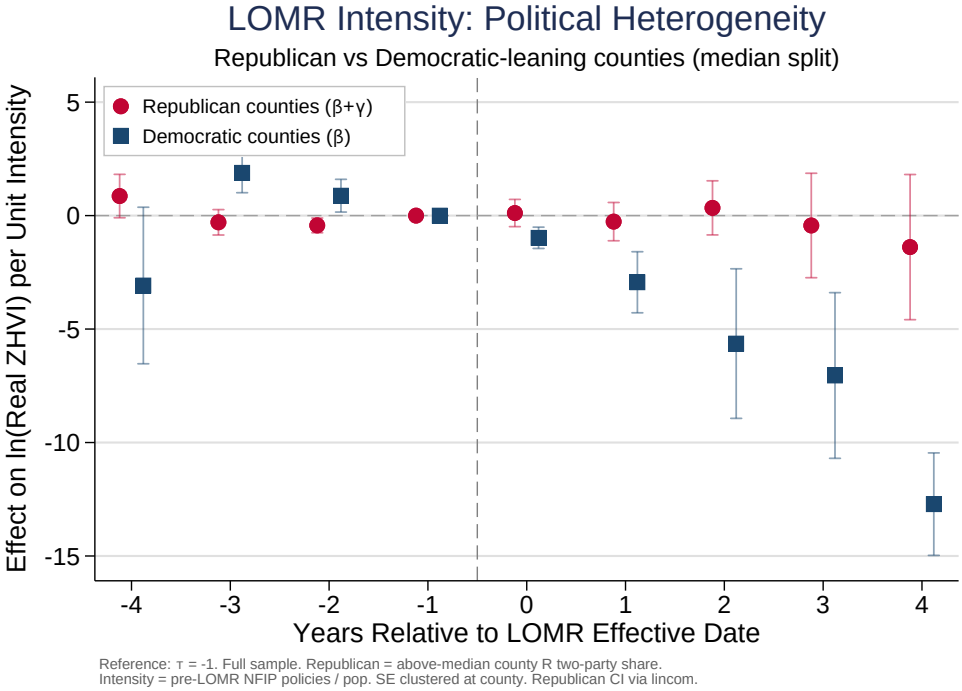
Republican = above-median county Republican two-party vote share.

Intensity = pre-LOMR NFIP policies / population.

Zip and county×year FE. SE clustered at county level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 7: Event Study by Political Lean



Note: Intensity-weighted event-study coefficients split by county-level Republican vote share (above/below median). Democratic-leaning baseline (solid) and Republican implied effect (dashed). Pre-treatment interaction F-test rejects parallel trends ($p < 0.001$); interpret as descriptive.

LOMR itself. I therefore present the partisan split as descriptive evidence that motivates further investigation rather than as a causal claim.

8.2 Robustness Checks

The Goodman-Bacon decomposition (Figure A6) is reassuring on weighting. About 95.4 percent of the identifying weight comes from never-treated versus treated timing comparisons, while only 0.3 percent comes from already-treated comparisons that are usually the main concern in staggered TWFE settings. However, the Callaway and Sant'Anna check (Figure A7) points to some tension with the baseline event study. The aggregated pre-treatment effect is positive and statistically different from zero, with $\text{Pre}_{avg} = 0.0059$ and $p = 0.002$, while the post-treatment average is imprecise, with $\text{Post}_{avg} = 0.0073$ and $p = 0.299$. I do not treat that estimator as a direct substitute for the baseline because it uses

annual data, a balanced panel, no controls, and no weights. Still, I do not want to ignore the discrepancy. The unemployment placebo (Figure A8) adds to that caution because the pre-trend test fails, with $F(3, 338) = 7.22$ and $p = 0.0001$.

Table 9 keeps ln(real ZHVI) as the dependent variable and changes the treatment specification across columns. Column 1 reproduces the weighted baseline event study, Column 2 removes the population weights, and Column 3 replaces the insurance-intensity interaction with a geographic-intensity interaction based on LOMR overlap area. I use the table to show which parts of the baseline result survive those alternative choices.

Excluding Florida reduces the late-post-treatment estimate from about -0.029 to -0.017 and removes statistical significance, which suggests that the results are most precise in states with heavier coastal LOMR activity. Clustering at the state level leaves the $\tau{+}4$ estimate negative, and the state-clustered 95 percent confidence interval remains below zero at $[-0.0545, -0.0029]$. Restricting the sample to ZIP codes with population of at least 1,000 leaves the baseline result essentially unchanged: the $\tau{+}4$ estimate is -0.0250 with a standard error of 0.0115 in a sample of 216,776 ZIP-quarter observations. Unweighted estimates are close to zero, and the geographic-intensity interaction based on LOMR overlap area is null and wrong-signed even though the policy-intensity interaction in Section 6.2 is negative and significant. That contrast points more toward salience than a simple area-overlap gradient, although the geographic measure is also a coarser proxy for the exposure that buyers and lenders actually face. Table A4 with Figure A5 shows that the non-stable SFHA-share proxy subsample remains negative but insignificant, with $\tau{+}4 = -0.019$ and a standard error of 0.012. Overall, these checks suggest that the baseline pattern is present but not universal across every alternative specification.

Table 9: ln(Real ZHVI) on LOMR (Robustness)

	(1) Main (weighted)	(2) Unweighted	(3) Geographic Intensity
τ = -4 (3+ yrs pre)	-0.00839	-0.00513	
	(0.00765)	(0.00583)	
τ = -3 (2-3 yrs pre)	-0.000353	-0.00113	
	(0.00470)	(0.00358)	
τ = -2 (1-2 yrs pre)	-0.00138	-0.00165	
	(0.00303)	(0.00231)	
τ = 0 (0-1 yr post)	-0.0000485	0.000768	
	(0.00231)	(0.00204)	
τ = +1 (1-2 yrs post)	-0.00454	-0.000575	
	(0.00400)	(0.00357)	
τ = +2 (2-3 yrs post)	-0.00926	-0.00257	
	(0.00599)	(0.00515)	
τ = +3 (3-4 yrs post)	-0.0139*	-0.00386	
	(0.00714)	(0.00639)	
τ = +4 (4+ yrs post)	-0.0287**	-0.00785	
	(0.0113)	(0.0120)	
geo_m4			-0.0751
			(0.0461)
geo_m3			-0.0201
			(0.0341)
geo_m2			-0.0329
			(0.0256)
geo_p0			0.0144
			(0.0209)
geo_p1			-0.000154
			(0.0318)
geo_p2			0.0135
			(0.0395)
geo_p3			0.0267
			(0.0520)
geo_p4			0.0536
			(0.0654)
Observations	228,002	228,002	228,002
Within R^2	0.0029	0.0014	0.0017

Standard errors in parentheses

Zip and county×year fixed effects.

Standard errors clustered at county level.

Geographic intensity = LOMR polygon area / ZCTA area.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

9 Discussion

9.1 Interpretation

Taken together, the results suggest that LOMRs matter less as immediate damage shocks and more as slow-moving regulatory information shocks. The baseline event study shows little movement on impact and a more negative price response several years later. The continuous intensity specification points to larger declines where pre-treatment NFIP penetration is higher, although the quartile pattern is non-monotonic. These themes converge in one interpretation. A map revision has more pricing content where more households, lenders, and insurers already sit near the margin where flood designation affects expected costs.

The weaker results still shape what I think the paper can and cannot say. I do not see strong average post-LOMR increases in NFIP policy counts, so the main home-value response does not look like a universal immediate insurance-take-up effect. Also, the significant policy-intensity interaction does not generalize to the geographic-overlap intensity measure, which is null in the robustness table. I take that difference as suggestive that economic salience matters more than simple overlap area, although the geographic measure is less tightly linked to the transaction margin. Disclosure-law interactions are weak, the upzoning and downzoning price splits are imprecise, and the political split fails the pre-trend standard. Therefore, I read the core result as gradual capitalization of revised official risk information, with supporting evidence on salience, but only mixed evidence on the exact transaction channel.

9.2 Limitations

Several limitations remain. The analysis uses ZIP-level panels rather than parcel-level or transaction-level data, so I cannot observe which homes actually cross the SFHA boundary

or how quickly lenders and buyers respond inside each ZIP. The baseline specification also pools revisions that may raise or lower mapped exposure because my ZIP-level treatment construction does not directly identify direction from the LOMR overlay alone. I proxy for directional heterogeneity with changes in NFIP SFHA share around the LOMR, but that measure is an exposure proxy rather than a direct parcel-level map transition. Therefore, the heterogeneity exercises should be read as suggestive rather than definitive.

Some auxiliary checks also reduce confidence in a simple causal reading. The main result is more precisely estimated when larger ZIP codes receive more weight, and it becomes weaker in unweighted specifications. The Callaway and Sant'Anna check does not reproduce the baseline pattern cleanly, the unemployment placebo shows problematic pre-trends, and the leave-one-out exercise shows meaningful sensitivity to Florida. I do not think these results erase the baseline event-study pattern, but they do narrow the claim. The evidence is consistent with gradual repricing after LOMRs, and I cannot rule out residual local trends tied to engineering review, development, or other local changes around the same time.

10 Conclusion

This paper asks whether official FEMA flood-map revisions affect local housing prices. Using a ZIP-by-quarter panel for coastal U.S. ZIP codes from 2009 to 2022 and a staggered event-study design, I find that first in-window LOMRs are associated with gradual declines in home values rather than an immediate one-period drop. In the baseline specification, the effect reaches about 2.8 percent four or more years after treatment. In the intensity specification, the implied effect at the sample mean insurance exposure is about 4.2 percent. Those results are consistent with the idea that map revisions matter most where flood risk is already economically salient.

Mechanism evidence on insurance take-up is limited, and the exploratory heterogeneity

results are mixed, so I do not claim a single channel with confidence. Still, the main pattern is consistent across the preferred specification. Official flood-risk reclassification can be capitalized into housing markets over time even without a contemporaneous flood. Future work with parcel-level transactions, lender data, and direct map-transition measures could separate information, insurance, and financing channels more cleanly.

Data Availability

Replication files, code, and the constructed analysis dataset are available at https://fear.camkeith.me.

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A Additional Tables

This appendix collects supplementary estimates that I discuss more briefly in the main text. I include them here because they help show what becomes weaker, more fragile, or less precisely estimated once I move away from the baseline event-study design.

Table A1 keeps ln(real ZHVI) as the dependent variable and splits the event-study coefficients into ZIP codes with proxy SFHA-share increases and decreases around the LOMR. The key regressors are therefore the same event-time indicators as in the baseline specification, estimated separately for the two proxy zoning directions. Neither path is individually precise. At $\tau+4$, the upzoned estimate is -0.0142 with a standard error of 0.0170, while the downzoned estimate is -0.0237 with a standard error of 0.0186. Those estimates remain statistically indistinguishable from zero, so I treat the sign pattern as descriptive rather than probative.

Table A2 also uses ln(real ZHVI) as the dependent variable, but the key regressors are event-time indicators interacted with state disclosure-law categories. The strict-disclosure category includes nine states with verified mandatory flood-zone disclosure on the seller form (CA, IL, IN, LA, MS, OR, SC, TX, WI), and the broad category adds four states with weaker or more recent requirements (FL, NC, NY, VA). The table asks whether the home-value response differs across these regimes. The late post-treatment interaction for the strict-disclosure specification is 0.0065 with a standard error of 0.0231, while the corresponding broad-disclosure interaction is -0.0260 with a standard error of 0.0238. Neither post-treatment interaction is statistically distinguishable from zero. However, the strict-disclosure lead at $\tau-4$ is positive, 0.0280 with a standard error of 0.0166, which is another reason to avoid a strong causal interpretation of these subsample differences.

Table A3 changes the dependent variables to insurance outcomes and estimates them separately for the proxy upzoning and downzoning groups. The key regressors remain the event-time indicators, but the table shows whether the insurance response differs with the direction of the proxy map-status change. The clearest pattern is on the policy side in downzoned ZIP codes. The coefficients are -0.0648 with a standard error of 0.0311 at $\tau+1$, and -0.0924 with a standard error of 0.0429 at $\tau+2$. Upzoned policy effects remain close to zero throughout. I read this asymmetry as suggestive evidence that the removal of mandatory coverage may matter more clearly than the imposition of coverage, although the broader mechanism evidence remains limited.

Table A4 returns to ln(real ZHVI) and restricts the baseline event study to ZIP codes with non-stable NFIP SFHA-share changes around the LOMR. The key regressors are still the event-time indicators, but the sample is narrower and closer to the subset where the proxy zoning-direction measure is most likely to reflect a real shift in mapped exposure. The late-post-treatment coefficient falls to -0.0192 with a standard error of 0.0124, which remains statistically insignificant. Therefore, the baseline result does not become sharper when I focus only on the subset most likely to capture meaningful map-status changes under the proxy definition.

Table A1: ln(Real ZHVI) on Upzoning vs Downzoning

	(1) Upzoned (Risk ↑)	(2) Downzoned (Risk ↓)
τ = -4 (3+ yrs pre)	-0.00612	
	(0.0115)	
τ = -3 (2-3 yrs pre)	0.00552	
	(0.00659)	
τ = -2 (1-2 yrs pre)	-0.00176	
	(0.00260)	
τ = 0 (0-1 yr post)	0.00441	
	(0.00355)	
τ = +1 (1-2 yrs post)	0.000825	
	(0.00633)	
τ = +2 (2-3 yrs post)	0.000427	
	(0.0102)	
τ = +3 (3-4 yrs post)	-0.000621	
	(0.0121)	
τ = +4 (4+ yrs post)	-0.0142	
	(0.0170)	
τ = -4 (3+ yrs pre)		-0.00462
		(0.0144)
τ = -3 (2-3 yrs pre)		-0.000508
		(0.0102)
τ = -2 (1-2 yrs pre)		-0.00215
		(0.00581)
τ = 0 (0-1 yr post)		0.000759
		(0.00621)
τ = +1 (1-2 yrs post)		-0.00167
		(0.00838)
τ = +2 (2-3 yrs post)		-0.00579
		(0.0122)
τ = +3 (3-4 yrs post)		-0.00851
		(0.0147)
τ = +4 (4+ yrs post)		-0.0237
		(0.0186)
Observations	208,170	205,208
Within R^2	0.0016	0.0018

Standard errors in parentheses

Zip and county×year FE. SE clustered at county level.

Each spec uses its treated subsample + all never-treated controls.

Risk direction classified by SFHA zone share change (±1pp) around LOMR.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A2: ln(Real ZHVI) on LOMR × Disclosure

	(1) Binary	(2) × Strict Disclosure	(3) × Broad Disclosure
τ = -4 (3+ yrs pre)	-0.00839	-0.0196**	-0.0136
	(0.00765)	(0.00846)	(0.0123)
τ = -3 (2-3 yrs pre)	-0.000353	-0.00280	-0.00556
	(0.00470)	(0.00438)	(0.00581)
τ = -2 (1-2 yrs pre)	-0.00138	-0.00172	-0.00319
	(0.00303)	(0.00238)	(0.00295)
τ = 0 (0-1 yr post)	-0.0000485	0.000317	0.00550*
	(0.00231)	(0.00272)	(0.00297)
τ = +1 (1-2 yrs post)	-0.00454	-0.00505	0.000766
	(0.00400)	(0.00512)	(0.00605)
τ = +2 (2-3 yrs post)	-0.00926	-0.00864	0.00157
	(0.00599)	(0.00837)	(0.00835)
τ = +3 (3-4 yrs post)	-0.0139*	-0.0126	0.00139
	(0.00714)	(0.0114)	(0.0117)
τ = +4 (4+ yrs post)	-0.0287**	-0.0324*	-0.00786
	(0.0113)	(0.0194)	(0.0199)
τ = -4 × Disclosure		0.0280*	
		(0.0166)	
τ = -3 × Disclosure		0.00468	
		(0.0107)	
τ = -2 × Disclosure		-0.000187	
		(0.00680)	
τ = 0 × Disclosure		-0.000305	
		(0.00453)	
τ = +1 × Disclosure		0.000922	
		(0.00807)	
τ = +2 × Disclosure		-0.00138	
		(0.0119)	
τ = +3 × Disclosure		-0.00269	
		(0.0139)	
τ = +4 × Disclosure		0.00645	
		(0.0231)	
τ = -4 × Broad Disc.			0.00840
			(0.0155)
τ = -3 × Broad Disc.			0.00768
			(0.00855)
τ = -2 × Broad Disc.			0.00276
			(0.00508)
τ = 0 × Broad Disc.			-0.00770*
			(0.00421)
τ = +1 × Broad Disc.			-0.00746
			(0.00792)
τ = +2 × Broad Disc.			-0.0149
			(0.0113)
τ = +3 × Broad Disc.			-0.0204
			(0.0145)
τ = +4 × Broad Disc.			-0.0260
			(0.0238)
Observations	228,002	228,002	228,002
Within R^2	0.0029	0.0036	0.0033

Standard errors in parentheses
Sample: all treated zips + never-treated controls (full event study sample).
ebin = baseline LOMR effect (non-disclosure). dbin/dbinb = differential disclosure effect.
Zip and county×year FE. SE clustered at county level.
Strict: CA, IL, IN, LA, MS, OR, SC, TX, WI (9 states).
Broad adds FL, VA, NC, NY (13 states).
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A3: ln(NFIP Policies) on Upzoning vs Downzoning

	(1) Policies (Up)	(2) Policies (Down)	(3) Claims (Up)	(4) Claims (Down)
up_m4	-0.0152		-0.0613**	
	(0.0341)		(0.0298)	
up_m3	-0.000135		-0.0839*	
	(0.0332)		(0.0450)	
up_m2	-0.0174		-0.0334	
	(0.0252)		(0.0389)	
up_p0	-0.0138		-0.0664**	
	(0.0127)		(0.0313)	
up_p1	-0.00873		-0.0448	
	(0.0161)		(0.0372)	
up_p2	0.00248		-0.0816**	
	(0.0244)		(0.0405)	
up_p3	-0.00130		-0.0490	
	(0.0293)		(0.0427)	
up_p4	-0.0102		-0.0693*	
	(0.0341)		(0.0388)	
down_m4		-0.0354		0.0227
		(0.0320)		(0.0170)
down_m3		-0.0560*		0.0599***
		(0.0306)		(0.0225)
down_m2		-0.0198		0.0577**
		(0.0253)		(0.0243)
down_p0		-0.0291		0.0965**
		(0.0323)		(0.0387)
down_p1		-0.0648**		0.0486*
		(0.0311)		(0.0290)
down_p2		-0.0924**		0.0513*
		(0.0429)		(0.0276)
down_p3		-0.0482		0.0291
		(0.0411)		(0.0239)
down_p4		-0.0649		0.0275
		(0.0698)		(0.0174)
Observations	208,170	205,208	208,170	205,208
Within R^2	0.0022	0.0025	0.0002	0.0002

Standard errors in parentheses

Zip and county×year FE. SE clustered at county level.

Each spec uses its treated subsample + all never-treated controls.

Risk direction: SFHA zone share Δ ($\pm$1pp threshold).

Reference period: τ = -1 (12-0 months before LOMR).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A4: ln(Real ZHVI) on SFHA-Crossing LOMRs

	(1) All LOMRs	(2) SFHA-Crossing Only
τ = -4 (3+ yrs pre)	-0.00839	-0.00625
	(0.00765)	(0.00907)
τ = -3 (2-3 yrs pre)	-0.000353	0.00335
	(0.00470)	(0.00588)
τ = -2 (1-2 yrs pre)	-0.00138	-0.00173
	(0.00303)	(0.00313)
τ = 0 (0-1 yr post)	-0.0000485	0.00249
	(0.00231)	(0.00349)
τ = +1 (1-2 yrs post)	-0.00454	-0.000532
	(0.00400)	(0.00498)
τ = +2 (2-3 yrs post)	-0.00926	-0.00295
	(0.00599)	(0.00732)
τ = +3 (3-4 yrs post)	-0.0139*	-0.00504
	(0.00714)	(0.00926)
τ = +4 (4+ yrs post)	-0.0287**	-0.0192
	(0.0113)	(0.0124)
Observations	228,002	215,580
Within R^2	0.0029	0.0020

Standard errors in parentheses

Zip and county×year FE. SE clustered at county level.

SFHA-crossing = zone_risk_direction is 'up' or 'down' (not 'stable').

Reference period: τ = -1 (12-0 months before LOMR).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A5 keeps ln(real ZHVI) as the dependent variable but replaces the full event-time path with one post-LOMR indicator. The key regressor is therefore a single treated-after-treatment dummy in a pooled two-way fixed-effects difference-in-differences design. The Post-LOMR coefficient is -0.0070 with a standard error of 0.0081. That estimate is much smaller than the late-period event-study coefficient and statistically indistinguishable from zero, which is exactly what I would expect if a pooled post indicator averages together early near-zero periods and later negative periods.

Table A5: ln(Real ZHVI) on LOMR (TWFE DiD)

	(1) TWFE DiD
Post-LOMR	-0.00699
	(0.00805)
County Unemp. Rate (	(0.000346)
NFIP Policies (qtr avg)	0.000000941
	(0.0000101)
Constant	12.91***
	(0.00219)
Observations	228,002
Within R^2	0.0016

Standard errors in parentheses

Zip and county×year FE. SE clustered at county level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

B Additional Figures

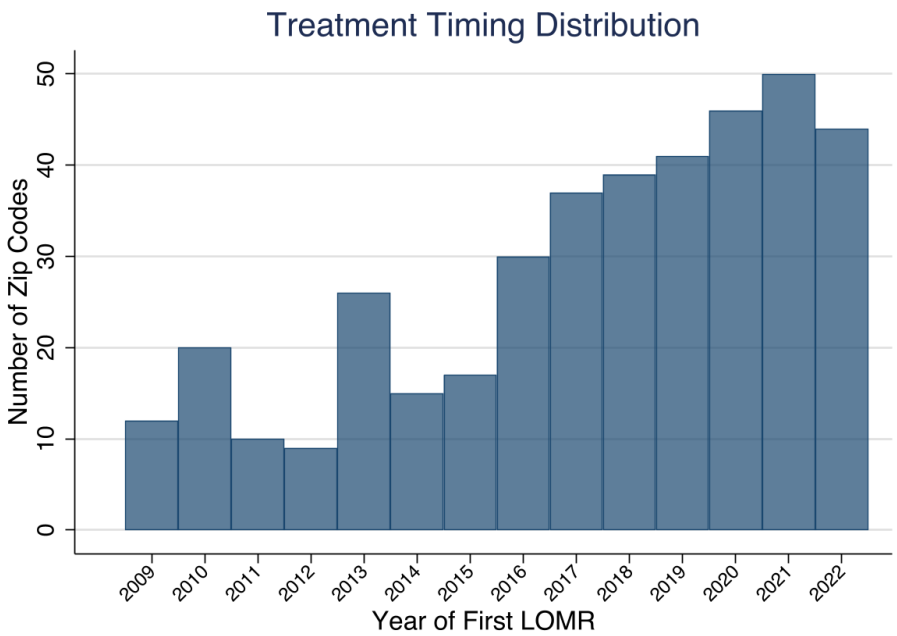
The appendix begins with the treatment-timing histogram from the main data section and then collects the supplementary heterogeneity and diagnostic figures. The main value of these figures is not to reveal a hidden sharp effect. It is to show either that treatment timing is spread across the sample period or that confidence intervals widen once the treated sample is split into narrower groups.

Figure A1 shows that first LOMR years are spread across the sample period rather than concentrated in one or two years. The next four figures visualize the same supplementary heterogeneity patterns from the appendix tables. Both proxy upzoning and proxy downzoning paths remain imprecise across event time, the disclosure interactions do not produce a clear separation in the post-treatment period, the limited insurance response is concentrated more in the downzoned policy series than in the upzoned series, and the non-stable SFHA-share proxy subsample remains negative in late event time but less precisely estimated than the full baseline sample.

Figure A5 reports the Goodman-Bacon decomposition for the simple TWFE design. The key detail is that 95.4 percent of the identifying weight comes from never-treated versus treated timing comparisons, while only 0.3 percent comes from already-treated comparisons. Therefore, the main concern in this decomposition is not extreme negative weighting from already-treated units. The larger issue is still that a single post indicator compresses the dynamic pattern.

Figure A6 reports the Callaway and Sant'Anna event study. The aggregated pre-treatment

Figure A1: Distribution of First LOMR Year

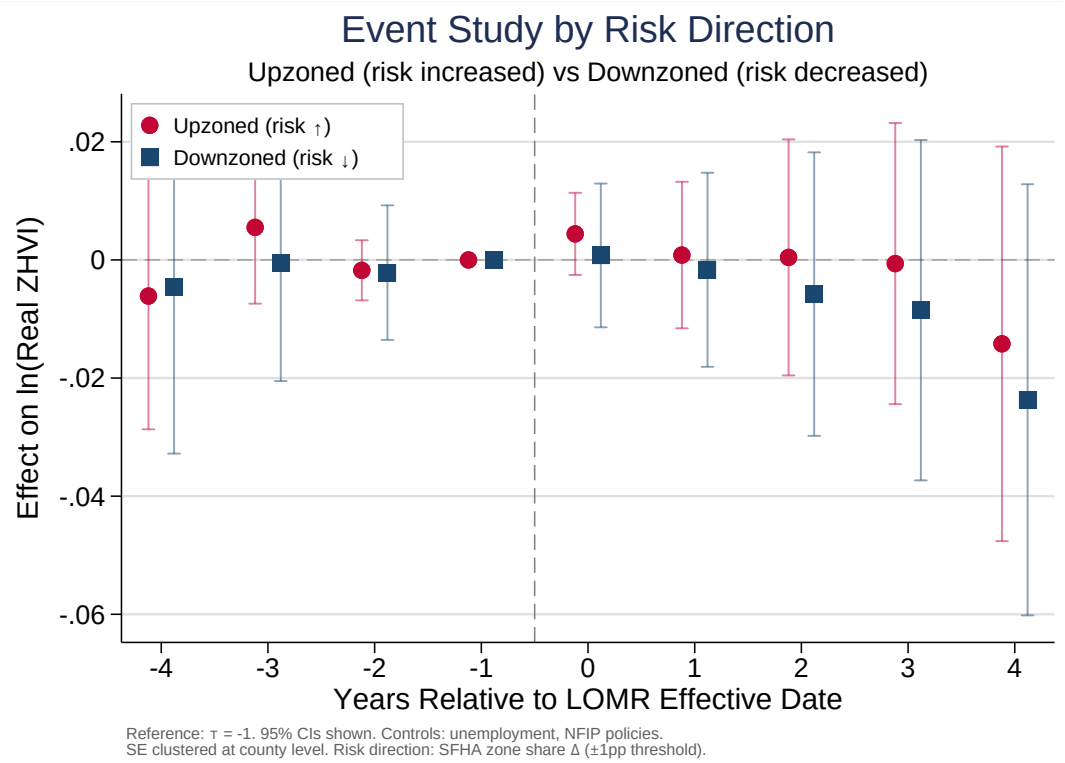


Note: Distribution of the first effective LOMR year across treated ZIP codes in the estimation sample (2009–2022).

average is 0.0059 with a standard error of 0.0020 and a p-value of 0.002. The post-treatment average is 0.0073 with a standard error of 0.0070 and a p-value of 0.299. That positive pre-treatment average is in tension with the main event-study design. Because this implementation uses annual data, a balanced panel, no weights, and no controls, I treat it as a cautionary check rather than the preferred estimate.

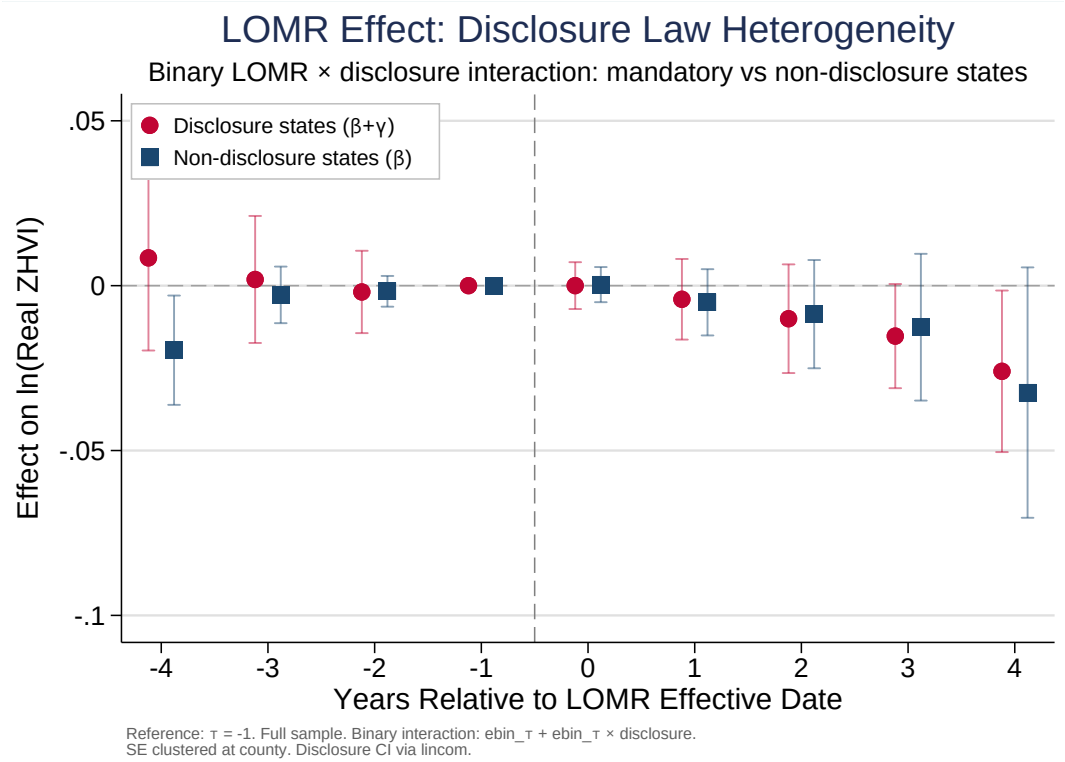
Figure A7 reports the unemployment placebo. The pre-trend test fails, with $F(3, 338) = 7.22$ and $p = 0.0001$, so this figure should be read as a warning sign rather than supportive evidence. I keep it in the appendix because it is informative about the possibility of residual local dynamics around treatment timing.

Figure A2: Event Study: SFHA-Share Increase vs. Decrease Proxy



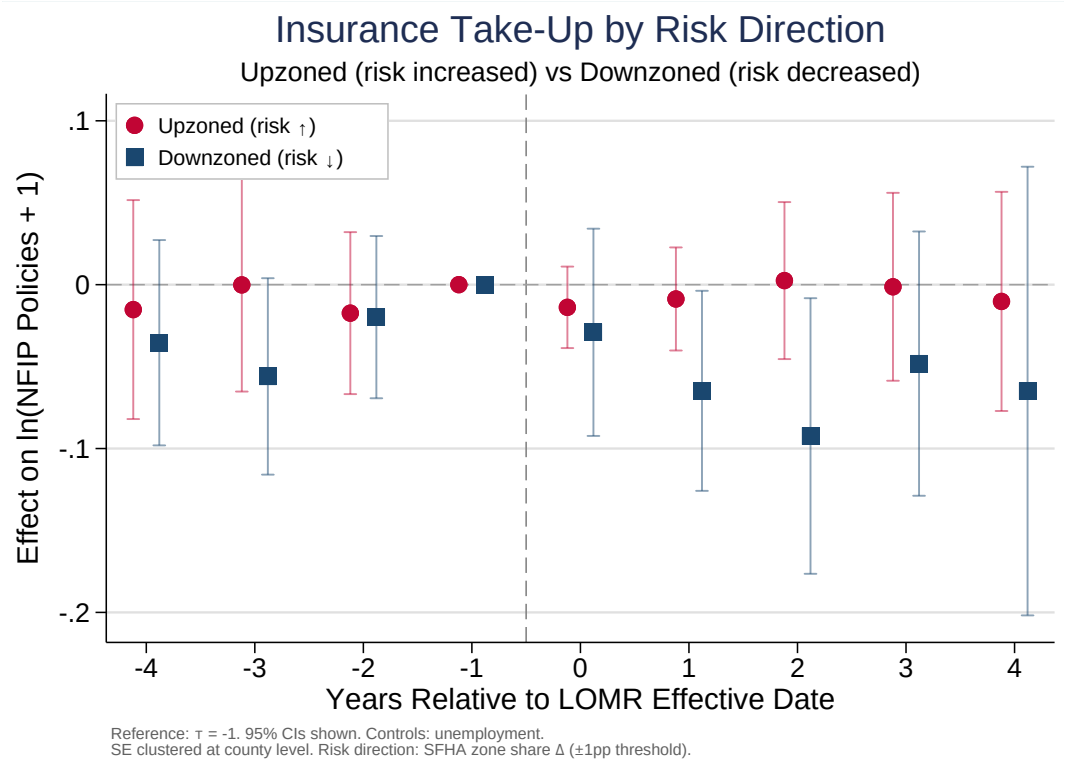
Note: Separate event-study coefficients for treated ZIP codes classified using increases or decreases in NFIP SFHA share around the LOMR.

Figure A3: Disclosure Law Heterogeneity Event Study



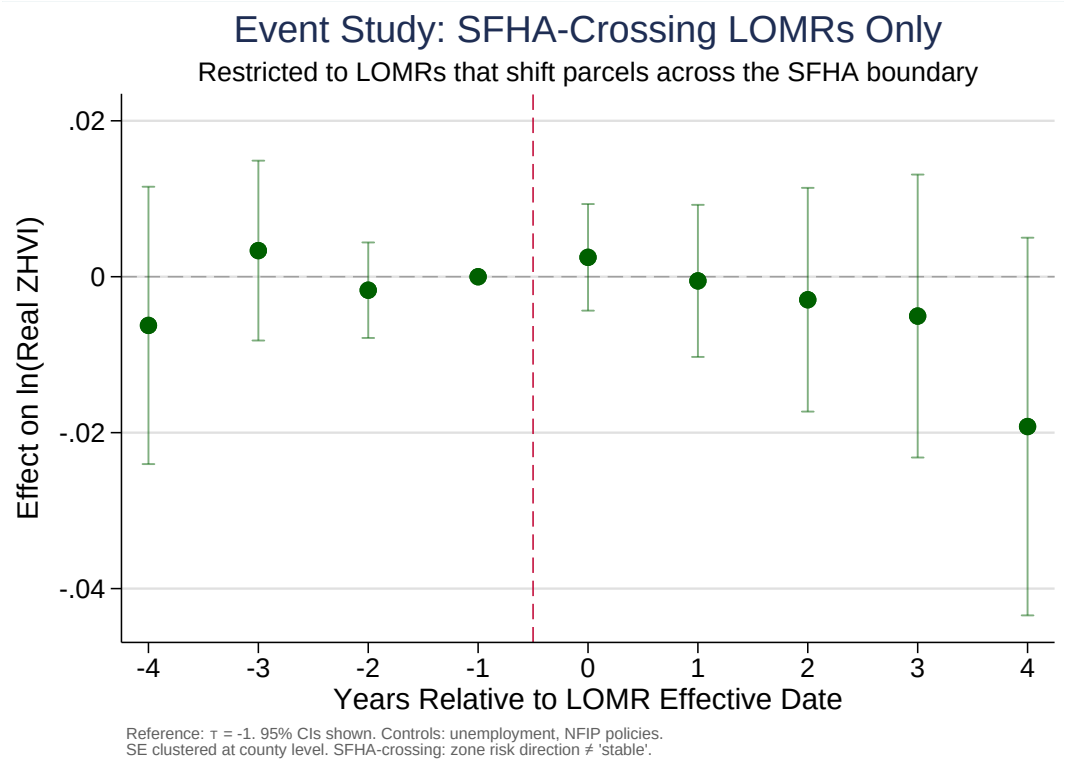
Note: Intensity-weighted event-study coefficients interacted with state mandatory flood disclosure indicator (9 strict states).

Figure A4: Insurance Policies Event Study by SFHA-Share-Change Proxy



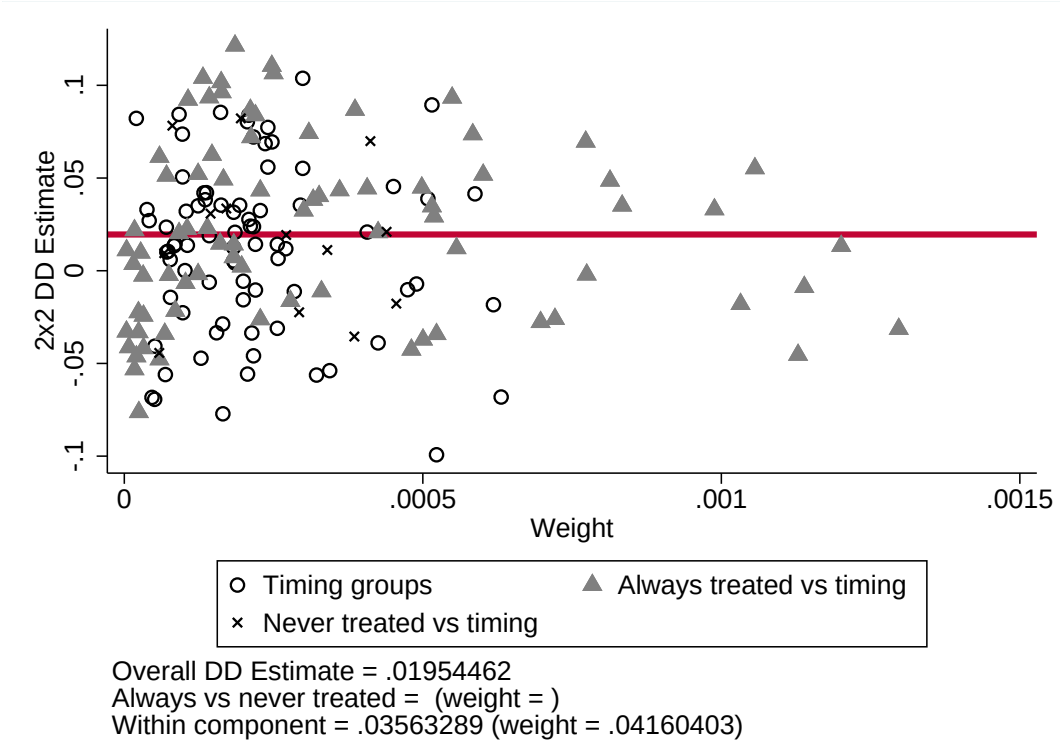
Note: Event-study coefficients for ln(NFIP policies + 1) estimated separately for treated ZIP codes classified by changes in NFIP SFHA share around the LOMR.

Figure A5: SFHA-Share-Change Proxy Subsample Event Study



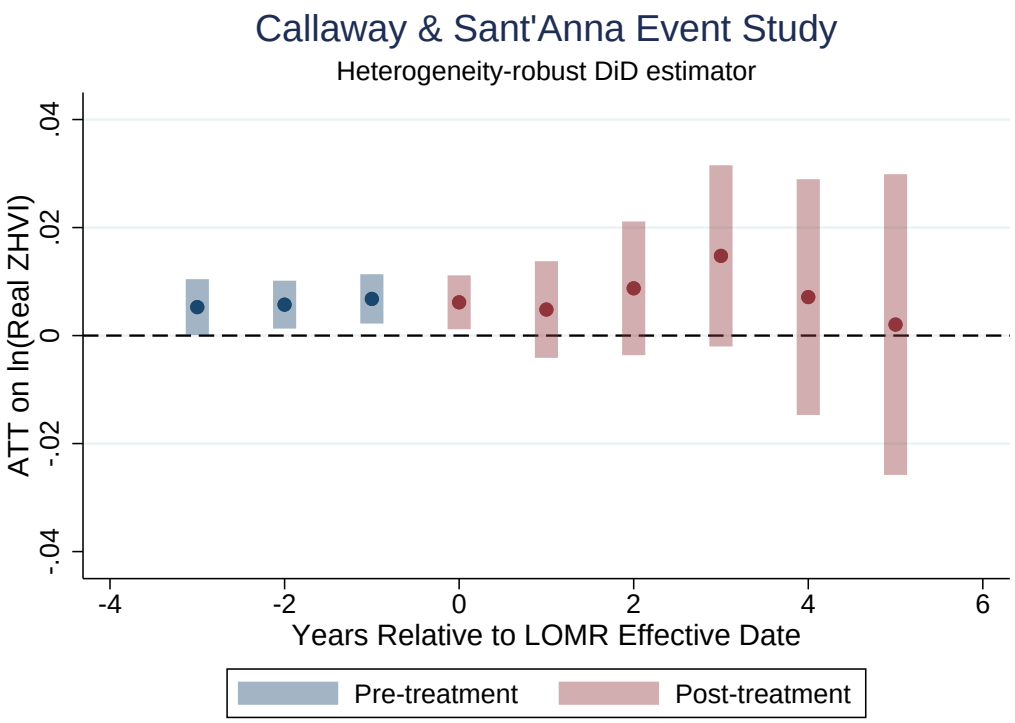
Note: Event-study coefficients restricted to LOMRs with non-stable NFIP SFHA-share changes around the LOMR. Same specification as Equation 1.

Figure A6: Goodman-Bacon Decomposition



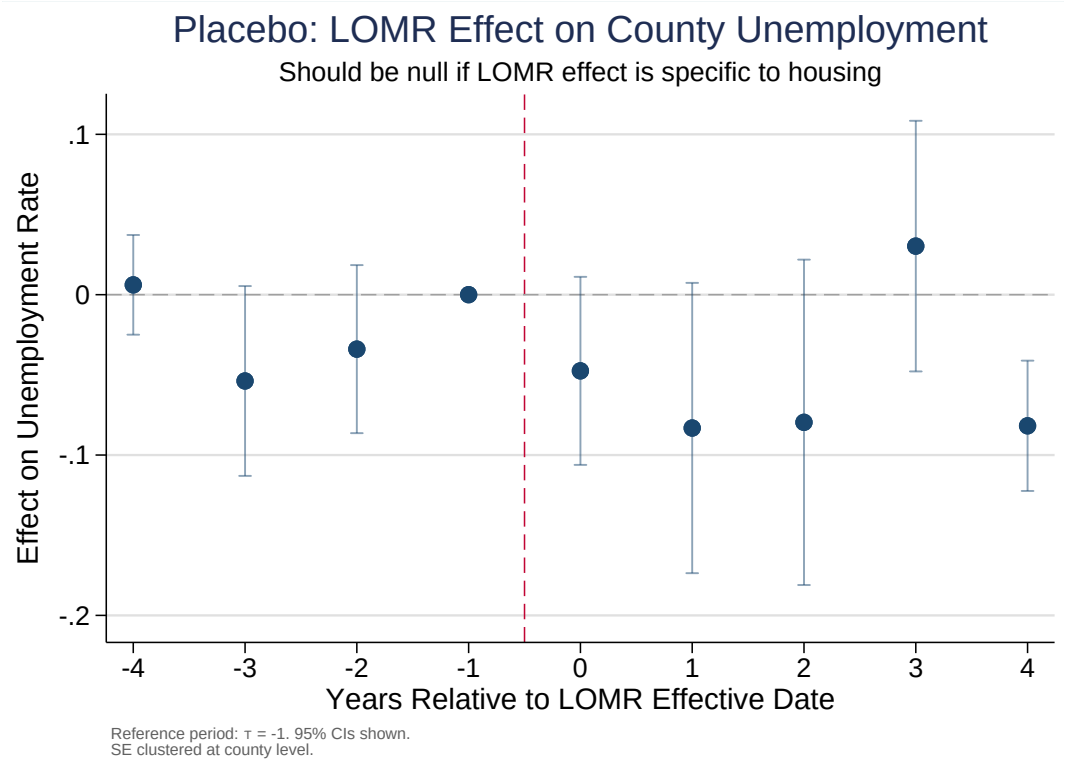
Note: Decomposition of the TWFE DiD coefficient into timing-group comparisons following Goodman-Bacon (2021). 95.4% of weight comes from never-treated vs. timing comparisons.

Figure A7: Callaway & Sant'Anna Event Study



Note: Callaway and Sant'Anna DiD estimator following Callaway and Sant'Anna (2021). Annual frequency, balanced panel, not-yet-treated and never-treated as controls. Cohorts restricted to 2009–2022.

Figure A8: Placebo Event Study: County Unemployment Rate



Note: Event-study coefficients with county unemployment rate as the dependent variable. Pre-trend F-test: $F(3, 338) = 7.22$, $p = 0.0001$.

Two additional diagnostics are not plotted here but still matter for interpretation. In the leave-one-out exercise, excluding Florida reduces the late-post-treatment estimate from about -0.029 to -0.017 and makes it statistically insignificant, while exclusions of other states leave the estimate close to the baseline value. Under state-level clustering, the $\tau+4$ coefficient remains -0.0287, and the standard error rises to 0.0132, which yields a 95 percent confidence interval of $[-0.0545, -0.0029]$. Those checks do not reverse the sign of the main result, and state-level clustering preserves statistical significance.

C Data Construction Details

I begin the geographic construction with 33,782 ZIP codes from the SimpleMaps ZIP database and restrict the sample to 5,450 ZIP codes located in NOAA coastal counties. Then, I overlay FEMA's effective S_LOMR polygons onto Census ZCTA boundaries to identify exposure to official map revisions. At the ZIP level, this procedure records the number of matched LOMRs, the first and last effective dates, and a geographic treatment-intensity measure equal to the overlap area between the LOMR polygon and the ZIP proxy geography divided by total ZIP area. In the broader coastal sample, the overlay yields 2,936 LOMR–ZIP pairs, 1,162 ever-treated ZIP codes, and 886 ZIP codes whose first observed LOMR falls inside the 2009–2022 window.

I merge those treatment measures to monthly Zillow ZIP-level home values, county unemployment rates from BLS LAUS, and OpenFEMA NFIP policies and claims aggregated to the ZIP-month level. The NFIP aggregation yields policy counts, total premiums, average premiums, SFHA zone share, claims counts, and total claims paid. I deflate Zillow values to December 2022 dollars with the Consumer Price Index for All Urban Consumers. The merged monthly panel is then exported for estimation, where it is collapsed to quarters (averaging monthly log values within each quarter) because the ZHVI is already a three-month smoothed index. At the estimation stage, I winsorize the quarterly log home values at the 1st and 99th percentiles and impose the sample restrictions: I drop ZIP codes treated before 2009 and ZIP codes with multiple LOMRs. The resulting quarterly estimation sample contains 4,272 ZIP codes.

The heterogeneity variables require additional construction choices. I define policy intensity as the ZIP's pre-treatment average NFIP policy count divided by population, and I drop ZIP codes with values above one because those cases imply implausible denominators. I classify proxy upzoning and downzoning by comparing mean NFIP SFHA zone share in the 12 months before and after the LOMR effective date; ZIP codes with a change exceeding ± 1 percentage point are classified as upzoned or downzoned, and the remainder as stable. This measure should be read as an exposure proxy rather than a direct parcel-level map transition. I construct the political heterogeneity measure from county-level mean Republican two-party vote share, split at the sample median, and I construct the disclosure variables from hand-coded state flood-disclosure rules. Those choices still shape the interpretation of the heterogeneity results, because each one trades cleaner coverage for a less direct measure of the underlying institutional channel.